NEUROHIRE: AI-DRIVEN RECRUITMENT BOOTH

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NeuroHire: AI-Driven Recruitment Booth

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NeuroHire: AI-Driven Recruitment Booth

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Dedication

We dedicate this project to our parents and families, whose unwavering support, patience, and encouragement have been the foundation of our academic journey. Their belief in us has continuously motivated us to strive for excellence and overcome every challenge along the way.

We extend our sincere gratitude to our mentors and instructors for their guidance, valuable insights, and continuous support throughout the development of this project. Their mentorship played a crucial role in shaping our understanding and helping us translate our ideas into a practical and meaningful system.

This project is especially dedicated to individuals seeking employment in mass hiring environments, particularly those with limited or no formal skills. Such candidates are often evaluated through processes that are inconsistent and time-consuming. NeuroHire is built with the aim of improving large-scale recruitment by introducing a more structured, efficient, and unbiased approach to candidate evaluation, ensuring that individuals are assessed based on their responses and behavior during the screening process.

We hope this work contributes toward improving hiring practices and supports organizations in making faster and more consistent recruitment decisions at scale.

Lastly, we acknowledge the dedication, collaboration, and perseverance of our team members, whose combined efforts made the successful completion of this project possible.

Acknowledgements

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We would also like to extend our appreciation to KFC for their support in providing us with relevant data and for facilitating user testing. Their cooperation allowed us to evaluate our system in a more practical and realistic setting, which significantly contributed to the effectiveness of our solution.

Lastly, we acknowledge the efforts of everyone who directly or indirectly contributed to the completion of this project.

Abstract

Recruitment at scale is difficult to perform consistently: large applicant pools, tight timelines, and fragmented tools for CVs, interviews, and notes often lead to uneven screening. NeuroHire addresses this gap as an integrated, AI-assisted recruitment support system that combines CV analysis, speech-based interview understanding, and behavioral video cues in a single workflow for high-volume hiring.

The platform supports a multi-role environment (candidates, recruiters, companies, and platform administration). Candidates choose an interview language (including English–Urdu bilingual support), upload a CV, select a job, and record structured video answers. CVs are captured with OCR and related parsing and scored against job descriptions; interview audio is transcribed and analyzed for answer relevance using large language models, together with clarity and confidence. Lightweight computer-vision analysis supplements spoken content with behavioral indicators such as gaze and head movement, while video quality checks flag unreliable segments for human review.

Module outputs are fused into weighted, explainable scores and recruiter-facing recommendations rather than opaque decisions. NeuroHire is designed to reduce repetitive manual review, improve comparability across applicants, and keep humans in the loop for oversight, appeals, and final judgment in mass-hiring settings.

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1. Introduction

1.1 Problem Statement

Recruitment and selection are critical processes for every organization, yet traditional hiring practices are often time-consuming, labor-intensive, and prone to inconsistency. Recruiters must manually review large numbers of CVs, shortlist candidates, conduct interviews, and compare applicants across multiple criteria. This becomes even more difficult in remote and high-volume hiring settings, where the number of applicants can exceed the capacity of human evaluators.

Recent advances in artificial intelligence have improved several parts of the recruitment pipeline, such as resume screening, interview automation, and candidate matching. However, many existing systems address only isolated tasks. Some focus mainly on resume parsing, others on speech recognition, while some analyze visual or behavioral cues from interviews. This fragmentation creates a gap in practical hiring systems, since recruiters often need a unified view of candidate suitability rather than separate outputs from disconnected tools.

Another challenge is that hiring decisions are not based only on qualifications written in a CV. Spoken responses, communication quality, clarity of answers, and observable behavioral cues during interviews may also provide useful information. Existing approaches often do not combine these dimensions in a structured and interpretable way. Moreover, concerns related to fairness, transparency, and reliability remain important when artificial intelligence is applied in recruitment.

Therefore, the problem addressed in this project is the lack of an integrated and explainable recruitment support system that can analyze candidate information from multiple modalities, including CVs, interview speech, and behavioral video cues, in a single framework.

1.2 Proposed Solution

To address this problem, this project proposes **NeuroHire**, an AI-assisted recruitment support system for multimodal candidate evaluation. The system is designed to assist recruiters by combining three major capabilities within a unified pipeline.

First, NeuroHire performs **CV analysis** to extract and organize relevant candidate information, enabling structured screening of resumes. This helps reduce the manual effort required in early-stage shortlisting and supports more consistent initial evaluation.

Second, the system performs **speech-based interview analysis**. Candidate responses from video interviews are transcribed and processed to assess their relevance to the given interview question, along with qualities such as clarity and confidence. This enables the system to move beyond simple transcription and support structured understanding of candidate responses.

Third, NeuroHire includes **behavioral video analysis** to observe cues such as gaze direction, head movement, and facial stability during interviews. These cues are not treated as final indicators on their own, but as supportive signals that can help assess engagement and interview interaction behavior.

The outputs of these modules are combined into a structured evaluation framework that supports decision-making for recruiters. In addition, the system includes quality checks so that poor video conditions, such as blur or inadequate lighting, can be flagged for human review. In this way, NeuroHire aims to provide a more comprehensive and practical recruitment aid than systems that focus on only one stage of hiring.

1.3 Intended User

NeuroHire is designed for a multi-user recruitment environment involving candidates, recruiters, companies, and a super admin. Each user type interacts with the system according to its specific role in the hiring process.

Candidate: The candidate is one of the primary users of the system. A candidate interacts with NeuroHire by selecting a preferred language for the interview, uploading a CV, selecting an available job, reviewing submitted information, and recording a video interview response. This allows the system to collect the required data for CV-based screening, speech analysis, and behavioral evaluation.

Recruiter: Recruiters use the system to manage recruitment activities related to their organization. Their responsibilities include adding, editing, and deleting job postings, reviewing applicants for posted jobs, and monitoring statistics related to jobs and applicants. Recruiters can also report issues or submit error-related complaints through the system when necessary.

Company: The company user supervises recruiter-level activity within the platform. A company can manage recruiters by accepting or rejecting recruiter registrations and can review overall statistics related to recruitment activity. This role helps ensure that only authorized recruiters are allowed to operate under the company's account.

Super Admin: The super admin is responsible for platform-level administration. This user manages companies by accepting or rejecting company registrations and can review system-wide statistics. The super admin ensures overall control, governance, and proper onboarding of organizations using the platform.

Overall, NeuroHire is intended for a structured hiring ecosystem in which each user type contributes to a different stage of the recruitment and administration process.

1.4 Project gantt chart and deliverables

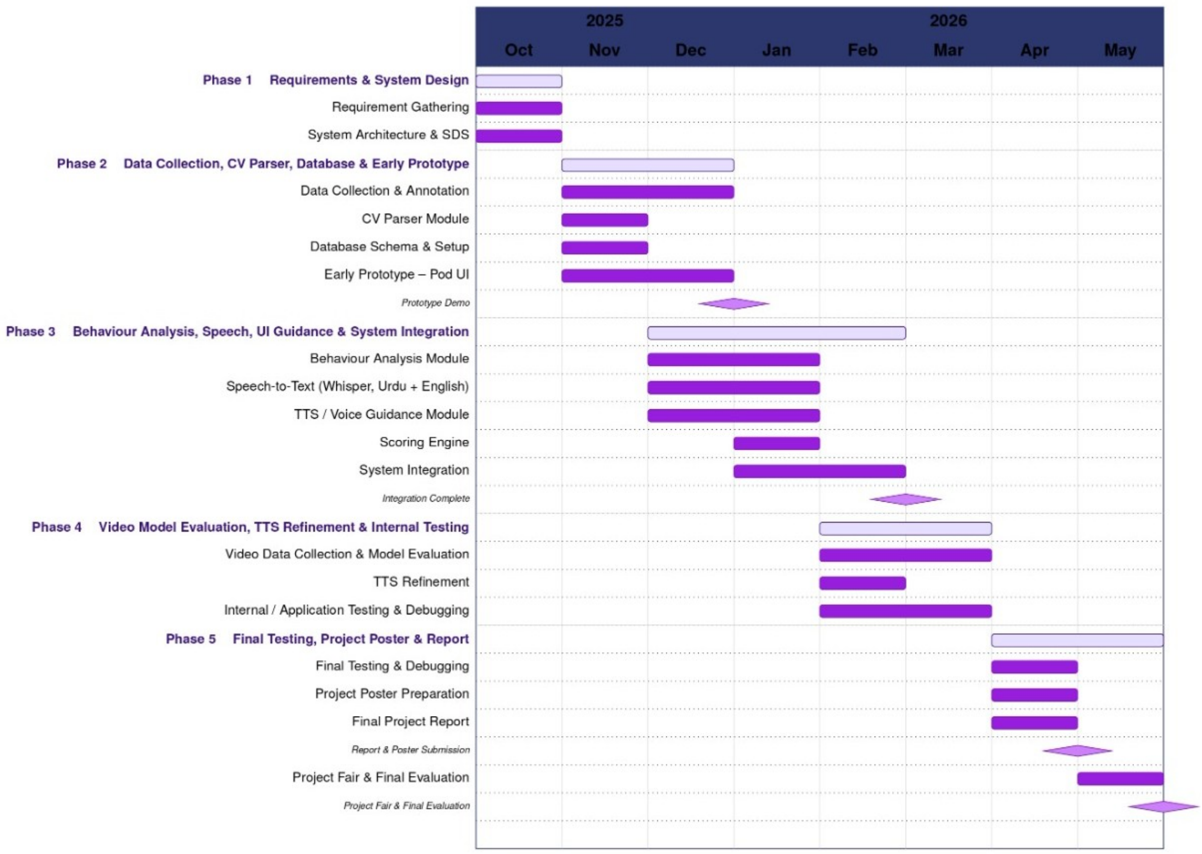

Figure 1.1: Project Timeline and Development Phases

1.5 Key Challenges

The development of NeuroHire involves several important challenges.

One major challenge is **multimodal integration**. NeuroHire combines CV analysis, speech-based response evaluation, and behavioral video analysis within a single system. Since these modules process different types of input data, their outputs must be integrated carefully so that the final evaluation remains meaningful, consistent, and easy to interpret. This challenge can be addressed by designing the system using a modular architecture, where each component performs its own task independently and then passes structured outputs to a central scoring or decision layer.

Another challenge is **speech transcription accuracy**. Since the system relies on transcription of interview responses before evaluating answer relevance, clarity, and confidence, any error in speech recognition may affect downstream analysis. This issue becomes more significant in multilingual settings, varied accents, or noisy environments. A possible way to address this challenge is to use a robust speech recognition model, support controlled interview conditions, and include preprocessing or confidence-based checks before passing the transcript to later stages of analysis.

The system also faces challenges in **behavioral cue interpretation**. Signals such as gaze direction, blinking, facial stability, or head movement can provide useful indicators of candidate engagement and interaction behavior, but they should not be treated as direct proof of candidate suitability or dishonesty. These cues may be influenced by environmental conditions, camera placement, internet quality, or natural individual differences in behavior. This challenge can be addressed by treating behavioral features as supportive indicators only, rather than final decision variables, and by allowing uncertain cases to be flagged for human review.

Another important challenge is **video quality and environmental variability**. Since NeuroHire depends on video input for behavioral analysis, poor lighting, blurred frames, low-resolution webcams, or unstable internet connections may reduce the reliability of extracted visual features. To address this issue, the system can include video quality checks such as blur detection and lighting assessment, and discard or down-weight unreliable behavioral scores when the input quality falls below acceptable thresholds.

A further challenge is **fairness and explainability**. AI-based recruitment systems may raise concerns if their outputs are difficult to understand or if users believe the system may introduce bias. This is especially important in hiring contexts, where trust and transparency are essential. A possible way to address this challenge is to design NeuroHire as a decision-support system rather than a fully autonomous decision-maker, and to provide structured score components that help recruiters understand how the system reached its evaluation.

Finally, **user trust and practical adoption** remain important challenges. Even if the system performs well technically, recruiters and organizations may hesitate to rely on it unless it aligns with real hiring workflows and produces understandable results. This challenge can be addressed by ensuring that the system output is simple, interpretable, and aligned with practical recruitment needs, while keeping human decision-makers involved in the final hiring process.

2. Literature Review

2.1 AI in Recruitment and Selection

Artificial intelligence has been widely explored as a means of improving recruitment workflows. Rathore discusses the impact of AI on recruitment and selection processes and explains that AI can automate repetitive tasks such as resume screening, interview scheduling, and candidate communication, thereby improving efficiency and reducing manual workload for recruiters [8]. The study also places these developments in the broader context of electronic human resource management, where digital platforms support online recruitment and virtual selection processes.

Similarly, Sha Ri examines the application of artificial intelligence in recruitment and selection, with emphasis on resume screening, interviewing, and competency assessment [9]. The study explains that natural language processing can be used to extract candidate information from resumes, while machine learning and data mining can support matching between candidates and job requirements. It further highlights the role of speech recognition and computer vision in analyzing candidate responses and non-verbal behavior during interviews.

These studies demonstrate that AI has become an important part of modern recruitment systems. However, they mainly discuss the role of AI at a general level and do not present a complete multimodal framework that combines CV analysis, interview response understanding, and behavioral evaluation in a single end-to-end system. This motivates the need for an integrated system such as NeuroHire.

2.2 Bias, Fairness, and Transparency in Algorithmic Hiring

As AI systems become more common in hiring, fairness and transparency have emerged as major concerns. Raghavan *et al.* analyze publicly available claims made by vendors of algorithmic hiring tools and show that many of these systems rely on question-based assessments, video interview analysis, and game-based evaluations [7]. Their study identifies significant shortcomings in transparency, especially regarding model validation and the fairness-related claims made by vendors.

A major concern raised in this work is that many algorithmic hiring systems depend on historical employee data. Because such data may reflect previous organizational bias, predictive models risk reproducing unfair patterns instead of reducing them [7]. The authors also argue that fairness is often treated in a limited way, for example through compliance-based measures, without addressing deeper issues in the data and modeling process.

This literature is important for NeuroHire because it highlights that hiring systems should not be treated as opaque decision-makers. Instead, they should provide structured and interpretable outputs. Unlike many black-box hiring tools criticized in prior work, NeuroHire is intended as a decision-support system that combines automated analysis with human-readable score components and the possibility of human review.

2.3 Speech Recognition for Interview Systems

Speech recognition is a foundational component of AI-assisted interview analysis. Radford *et al.* propose a large-scale weakly supervised speech recognition approach trained on approximately 680,000 hours of multilingual and multitask audio data [6]. Their work demonstrates strong robustness and generalization across multiple benchmarks and shows that performance improves significantly with increasing training scale and diversity. The study also supports multilingual transcription and translation, which is especially relevant for interview settings where candidates may respond in different languages.

This work is highly relevant to NeuroHire because accurate transcription is necessary before candidate responses can be analyzed for relevance, clarity, and confidence. However, the role of speech recognition in NeuroHire goes beyond general transcription. While Radford *et al.* focus on robust speech recognition itself, NeuroHire uses transcription as an enabling step for structured interview response evaluation in a recruitment context.

2.4 Multimodal Interview Assessment

A growing body of research supports the use of multimodal AI for automated interview evaluation. In *Automated Analysis and Prediction of Job Interview Performance*, the authors propose a framework for quantifying verbal and non-verbal behaviors in interview settings using prosodic, lexical, and facial features [2]. Their results show that higher speaking rate, fewer filler words, and more positive language are associated with better interview performance. The study also demonstrates that multimodal analysis outperforms single-modality approaches in predicting interview-related traits.

Nguyen *et al.* study body communication cues in employment interviews and show that gestures, posture, and speaking status can be used to predict hirability and personality-related outcomes [3]. Their work emphasizes that non-verbal behavior is closely linked with speech and contributes significantly to interview outcomes. However, the study also notes that part of the analysis depends on manual annotations, which limits full automation.

Zhang *et al.* further extend this area by investigating asynchronous video interviews and proposing a multimodal framework for assessing personality traits and interview performance using visual, audio, and verbal signals [10]. Their work addresses limitations in earlier studies by using a structured interview dataset designed for realistic job application scenarios and annotated by trained evaluators. The proposed baseline combines visual, audio, and textual representations and shows that multimodal systems can improve prediction performance in interview assessment tasks.

These studies strongly support the use of multimodal signals in video interview analysis. However, most of them focus on predictive modeling, dataset creation, or benchmarking rather than developing a practical recruitment support system. NeuroHire differs by using multimodal analysis within a larger system that also includes CV analysis, job-specific screening, and structured scoring for deployment-oriented candidate evaluation.

2.5 Behavioral Analysis

Behavioral signals such as gaze direction, blinking, and head movement are often used as indicators of attention and interaction behavior. Pimplaskar *et al.* propose a real-time eye blinking detection and tracking system using OpenCV and show that eye-related visual cues can be extracted effectively under real-time conditions [5]. Their work demonstrates the practical feasibility of monitoring eye activity using computer vision methods and highlights the broader relevance of eye-based cues for attention-related analysis.

Head pose is another important visual signal. Huttunen *et al.* study computer vision methods for head pose estimation and show that machine learning approaches can achieve strong performance in predicting yaw and pitch from facial images despite challenges such as limited data, illumination changes, and label ambiguity [1]. Their work provides technical support for using head orientation as a measurable behavioral feature.

For NeuroHire, these studies justify the use of gaze-related and head-pose-related signals as part of behavioral analysis during interviews. At the same time, such cues must be interpreted carefully. In NeuroHire, they are intended to act as supportive indicators of engagement and interaction behavior rather than standalone measures of competence or honesty.

2.6 Existing AI-Based Interview Systems

Some prior systems are relatively close to the goals of NeuroHire. Pandey *et al.* propose an AI-based interview system that combines speech recognition, facial expression analysis, resume parsing, and machine learning-based evaluation [4]. Their system uses natural language processing for content-related analysis, OpenCV and convolutional neural networks for visual analysis, and a composite scoring mechanism to generate overall interview performance scores.

This work demonstrates that combining textual, vocal, and visual analysis in recruitment systems is both feasible and useful. However, NeuroHire differs in several ways. It is designed as a more structured multimodal recruitment framework rather than a collection of loosely connected modules. In addition, NeuroHire places stronger emphasis on explainable scoring, system-level integration, and controlled use of behavioral cues within a broader recruitment support pipeline.

2.7 Discussion and Relevance to NeuroHire

The literature reviewed in this chapter shows that substantial progress has been made in AI-based recruitment, speech recognition, multimodal interview analysis, and behavioral video assessment. Prior work supports the use of AI for resume screening and recruitment automation [8, 9], robust multilingual transcription [6], multimodal prediction of interview performance [2, 3, 10], and extraction of visual behavioral cues such as eye movement and head pose [5, 1]. At the same time, literature on algorithmic hiring highlights concerns related to fairness, transparency, and interpretability [7].

A key observation from the reviewed work is that many existing approaches address only one part of the hiring process at a time. Some focus on recruitment automation at a high level, some on transcription, and others on multimodal interview prediction or behavioral feature extraction. Comparatively fewer systems attempt to combine CV analysis, speech-based answer evaluation, behavioral analysis, and structured decision support in one coherent framework.

This is where NeuroHire establishes its contribution. Rather than treating candidate evaluation as a single prediction task, NeuroHire integrates multiple modalities into a unified recruitment support pipeline. It combines CV-based screening, speech transcription and response analysis, and behavioral signal extraction from interview videos, while also emphasizing explainable outputs and practical usability. In this way, NeuroHire builds upon the strengths of prior work while addressing the lack of integration among existing approaches.

3. Software Requirement Specification (SRS)

This chapter provides detailed specifications of the system under development.

3.1 Functional Requirements

This section describes the functional requirements of the NeuroHire system, organized by user role as reflected in the system's use case diagrams. Each requirement describes a discrete, implementable behavior that the system must support.

- **Candidate Module:**
  - **FR-C1 – Language Selection:** The system shall allow a candidate to select their preferred interface language (Urdu or English) at the start of every session. All subsequent on-screen text, audio prompts, and guidance shall be rendered in the selected language.
  - **FR-C2 – Voice Guidance:** The system shall provide an optional voice guidance mode in which all instructions and prompts are delivered through audio output, enabling candidates with limited literacy to navigate the application process independently.
  - **FR-C3 – View Job Listings:** The system shall display all active job postings assigned to the current recruitment pod, allowing a candidate to browse available positions and select one before beginning their application.
  - **FR-C4 – CV Upload:** The system shall capture a candidate's curriculum vitae by scanning a physical document and store it in the database. The stored CV will be linked to the candidate's application record.
  - **FR-C5 – Review Extracted CV Information:** Upon successful CV capture, the system shall perform OCR-based text extraction and present the parsed information to the candidate for review. The candidate shall be able to correct any inaccuracies in the extracted data before submission.
  - **FR-C6 – Record Video Interview:** The system shall guide the candidate through a structured video recording session comprising job-specific screening questions, and shall start uploading the individual video response as soon the candidate clicks on submit.
    * **FR-C6a – Answer Screening Questions:** During the recording session, the system shall present each job-specific screening question defined by the

recruiter, prompting the candidate to respond on camera before proceeding to the next question.

  * **FR-C6b – Upload Video Responses:** The system shall upload the completed video responses to backend storage upon as soon as the cadidate clicks on submit. A progress indicator shall inform the candidate of upload status and provide a retry option in case of failure.

- **FR-C7 – Receive Submission Confirmation:** Upon successful submission of the CV and video responses, the system shall display a confirmation message acknowledging that the application has been received and is under review.

- **Recruiter Module:**

  - **FR-R1 – Login and Signup:** The system shall provide login and signup functionality for recruiters. During login, the system shall verify the entered credentials against the password hash stored in the database. Access to the dashboard shall only be granted after successful credential verification. New recruiter accounts shall remain in a pending state until approved by the associated Company user.
  - **FR-R2 – Post Vacancies:** The system shall allow recruiters to create new job postings, each comprising a job description and a set of screening questions configured by the recruiter.
    * **FR-R2a – Add Job Description:** When creating a vacancy, the system shall allow the recruiter to compose and save a detailed job description including role requirements.
    * **FR-R2b – Add Screening Questions:** The system shall allow recruiters to define a list of video-based screening questions for each job posting, which shall be presented to candidates during the video recording stage at the pod.
    * **FR-R2c – Edit Job Posting:** The system shall allow recruiters to edit any field of an existing job posting, including description, and screening questions. Changes shall take effect immediately for future applications.
    * **FR-R2d – Activate and Deactivate Jobs:** The system shall allow recruiters to toggle a job posting's status between active and inactive. Inactive jobs shall not be displayed to candidates at the pod and shall not accept new applications.
    * **FR-R2e – Delete Job Posting:** The system shall allow recruiters to permanently remove a job posting from active views via a soft-delete, retaining associated application records and AI assessments for audit.
    * **FR-R2f – Configure Scoring Weightage:** The system shall allow recruiters to define the relative weightage of CV-based evaluation and video-based evaluation for each job. The composite score of a candidate shall be calculated using these weightages to determine their ranking in the applicant list.
  - **FR-R3 – View Applicants:** The system shall present recruiters with a list of candidates who have applied for a specific job.

    * **FR-R3a – View CV:** The system shall provide access to the parsed CV data and the original uploaded CV document for each candidate within the applicant view.
    * **FR-R3b – View Video Interviews:** The system shall allow recruiters to play back recorded video responses submitted by a candidate directly within the dashboard.
    * **FR-R3c – View AI Recommendation:** The system shall display the AI-generated composite score and recommendation (Accept, Reject, or Send to Human Interview) for each candidate application.
  - **FR-R4 – Make Hiring Decision:** The system shall allow recruiters to record a final hiring decision for each applicant, which may agree with or override the AI recommendation. The decision shall be persisted in the recruiter decisions log.
    * **FR-R4a – Accept Candidate:** The system shall allow a recruiter to mark a candidate as accepted, updating the application status accordingly.
    * **FR-R4b – Reject Candidate:** The system shall allow a recruiter to reject a candidate, recording the decision and any associated comments in the recruiter decisions log.
    * **FR-R4c – Send to Human Interview:** The system shall allow a recruiter to forward a candidate to a human interview stage when further evaluation is required beyond the automated AI assessment.
  - **FR-R5 – Report System Errors:** The system shall provide a mechanism for recruiters to report technical issues or system errors encountered during use. Reports shall be logged and made available for Super Admin review.
  - **FR-R6 – Edit Recruiter Profile:** The system shall allow a recruiter to update their own account details, including full name, email address, and password. Password changes shall require confirmation of the current password and shall enforce minimum strength criteria.

- **Company / Head of HR Module:**
  - **FR-CO0 – Company Signup and Login:** The system shall provide secure account registration and login for Company users. A new company registration shall remain in a pending state until approved by the Super Admin. Login shall enforce authentication and session management consistent with other roles.
  - **FR-CO1 – Manage Recruiters:** The system shall allow the Company to manage all recruiter accounts associated with their organization, including the ability to approve, disable, and delete recruiter accounts.
    * **FR-CO1a – Approve Recruiters:** The system shall allow the Company to approve a recruiter account, granting that recruiter access to the dashboard and the ability to manage job postings.
    * **FR-CO1b – Disable Recruiters:** The system shall allow the Company to temporarily disable a recruiter account, preventing access without permanently removing the account or its data.

    * **FR-CO1c – Delete Recruiters:** The system shall allow the Company to permanently delete a recruiter account and its associated data from the organization.
  - **FR-CO2 – View All Jobs:** The system shall provide the Company with a consolidated view of all active and inactive job postings created by recruiters within their organization.
  - **FR-CO3 – View Applicants Overview:** The system shall provide the Company with an aggregated overview of applicant activity across all job postings, allowing high-level visibility into recruitment progress.
  - **FR-CO4 – Report an Issue:** The system shall provide the Company with a mechanism to report operational or technical issues to the Super Admin for resolution and tracking.
  - **FR-CO5 – Update Company Profile:** The system shall allow the Company user to update their organization's profile information, including company name, logo, and primary contact details. Logo changes shall be reflected in booth branding within one synchronisation cycle.

- **Super Admin Module:**
  - **FR-SA1 – View Platform Overview:** The system shall provide the Super Admin with a high-level dashboard displaying key operational metrics and the current health status of the platform, including total registered companies, active recruiters, and applications processed.
  - **FR-SA2 – View Recruiter Error Log:** The system shall allow the Super Admin to access a log of all system errors reported by or associated with recruiter activity.
  - **FR-SA3 – View Company Error Log:** The system shall allow the Super Admin to access a log of all system errors reported by or associated with company-level accounts.
  - **FR-SA4 – Change Error Status:** The system shall allow the Super Admin to update the resolution status of any logged error, marking it as either Resolved or Unresolved to track issue management progress.
  - **FR-SA5 – Manage Companies:** The system shall allow the Super Admin to manage all company accounts registered on the platform, including the ability to approve, reject, and delete company accounts.
    * **FR-SA5a – Approve Company:** The system shall allow the Super Admin to approve a company registration request, granting that company access to the NeuroHire platform and enabling its recruiters to begin operations.
    * **FR-SA5b – Reject Company:** The system shall allow the Super Admin to reject a company registration request, preventing the company from accessing the platform.
    * **FR-SA5c – Delete Company:** The system shall allow the Super Admin to permanently delete a company account and all associated recruiter accounts and data from the platform.

3.1.1 User Stories

The following table presents the user stories derived from the system use case diagrams, mapped to their corresponding functional requirements. Stories are organized into four epics aligned with the four human roles in the system. The **FR Ref.** column cross-references each story to the functional requirement(s) it realises, ensuring full traceability between stakeholder needs and system behavior.

Table 3.1: NeuroHire – User Story Table (32 Stories, 4 Epics)

US ID	Epic	Role	User Type	User Story	Acceptance Criteria	Module
EPIC-1	**Candidate Onboarding & Submission**					
US-01	EPIC-1	Candidate	Walk-in applicant	As a candidate, I want to select my preferred language (Urdu or English) at the start of my session so that all instructions are presented in a language I understand.	Language selector is the first screen at session start; applies to all on-screen text, audio guidance, and STT processing; can be changed before submission is confirmed.	Candidate Pod Interface
US-02	EPIC-1	Candidate	Walk-in applicant	As a candidate, I want to enable voice guidance so that I can follow the booth instructions without relying solely on reading the screen.	A voice-guidance toggle is available on the language selection screen; when enabled, all instructions and prompts are read aloud in the selected language throughout the session.	Candidate Pod Interface
US-03	EPIC-1	Candidate	Walk-in applicant	As a candidate, I want to view all job listings at the booth so that I can choose the role I want to apply for.	Pod displays all active jobs mapped to that booth; candidate can scroll, read descriptions, and confirm a selection before proceeding.	Candidate Pod Interface
US-04	EPIC-1	Candidate	Walk-in applicant	As a candidate, I want to scan my CV at the booth so that my qualifications are captured without needing a digital file.	CV image is captured, auto-enhanced for OCR readability, and stored in `CV_Data`; success confirmation shown before the next step.	CV Capture & OCR
US-05	EPIC-1	Candidate	Walk-in applicant	As a candidate, I want to review and edit the information extracted from my CV so that any OCR errors are corrected before submission.	Parsed fields (name,phone number, email, address) displayed in an editable form after OCR; confirmed values saved back to database before proceeding.	CV Capture & OCR
US-06	EPIC-1	Candidate	Walk-in applicant	As a candidate, I want to record video responses to job-specific screening questions so that I can demonstrate my communication and soft skills.	System presents each screening question in sequence; candidate records a response per question; video meets minimum length; stored in database; candidate may re-record once per question.	Video Capture Module
US-07	EPIC-1	Candidate	Walk-in applicant	As a candidate, I want my video responses to be uploaded and receive a submission confirmation so that I know my application has been successfully received.	Video files uploaded; confirmation screen shows a thank you message;.	Video Capture Module
EPIC-2	**Recruiter Workflow**					
US-08	EPIC-2	Recruiter	Recruiter	As a recruiter, I want to sign up and log in securely so that my account is linked to my organization and candidate data is protected.	Sign-up requires company name; account created with `status = pending` until approved by the Company user; login enforces password verification;	Auth & Account Mgmt
US-09	EPIC-2	Recruiter	Recruiter	As a recruiter, I want to create a new job posting with a description, and screening questions so that candidates are evaluated against the right criteria.	Job form saves title, description, and at least one screening question to database; job defaults to `inactive` until explicitly activated.	Job Management

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Table 3.1 – *continued from previous page*

US ID	Epic	Role	User Type	User Story	Acceptance Criteria	Module
US-10	EPIC-2	Recruiter	Recruiter	As a recruiter, I want to edit an existing job posting so that I can correct or update requirements without creating a duplicate.	Edit form pre-populates current values; changes saved immediately for future applications; existing applications retain a snapshot of original requirements.	Job Management
US-11	EPIC-2	Recruiter	Recruiter	As a recruiter, I want to activate or deactivate a job so that the pod only shows roles currently open for applications.	Toggle sets `job.status` to `active` or `inactive`; inactive jobs disappear from the pod list within one sync cycle; action written to `Recruiter_Logs`.	Job Management
US-12	EPIC-2	Recruiter	Recruiter	As a recruiter, I want to delete a job posting so that obsolete roles are permanently removed from active views.	Everything associated with the job is removed from the database.	Job Management
US-13	EPIC-2	Recruiter	Recruiter	As a recruiter, I want to configure the weightage of CV and video evaluations for a job so that candidate scores are calculated based on my preferred criteria.	Weightage for CV and video scores is saved per job; composite score is calculated accordingly; weightage can be updated at any time and takes effect immediately.	Job Management
US-14	EPIC-2	Recruiter	Recruiter	As a recruiter, I want to view all applicants for a job so that I can review who has applied before making a hiring decision.	Applicant list is displayed for the selected job; each applicant row shows basic candidate details and application status.	Recruiter Dashboard
US-15	EPIC-2	Recruiter	Recruiter	As a recruiter, I want to view an applicant's CV summary so that I can understand their qualifications and experience.	Parsed CV summary is displayed for each applicant; recruiter can access the uploaded CV for detailed review.	Recruiter Dashboard
US-16	EPIC-2	Recruiter	Recruiter	As a recruiter, I want to access an applicant's video interview so that I can review their recorded responses.	Video playback link is available for each applicant; recruiter can open and watch the submitted video interview.	Recruiter Dashboard
US-17	EPIC-2	Recruiter	Recruiter	As a recruiter, I want to view the AI recommendation for each applicant so that I can use system-generated support while making a decision.	AI recommendation is displayed for each applicant; recommendation is shown with relevant score or reasoning where available.	Recruiter Dashboard
US-18	EPIC-2	Recruiter	Recruiter	As a recruiter, I want to accept, reject, or forward a candidate to a human interview so that a formal decision is recorded for every application.	Action buttons visible per application row; outcome stored in `Recruiter_Decisions` with recruiter ID, timestamp, and optional override flag; action logged.	Recruiter Decisions
US-19	EPIC-2	Recruiter	Recruiter	As a recruiter, I want to edit my own profile (name, email, password) so that my account details remain accurate.	Profile edit form validates email format and password strength; confirmation email sent on save; old password required to set a new one.	Auth & Account Mgmt
US-20	EPIC-2	Recruiter	Recruiter	As a recruiter, I want to report a system error or unexpected behavior so that technical issues are escalated to the Super Admin.	"Report Issue" form captures error description and optional screenshot; creates an `Error_Log` entry tagged with `recruiter_id`; Super Admin is notified automatically.	Error Reporting
EPIC-3 Company (Head of HR) Management						
US-21	EPIC-3	Company	Head of HR / Company rep	As a Company user, I want to sign up and log in securely so that our organization can access NeuroHire and our data is protected.	Sign-up captures company name, industry, and primary contact; account set to `pending` until Super Admin approves; login enforces password authentication; failed attempts are logged.	Auth & Account Mgmt

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Table 3.1 – *continued from previous page*

US ID	Epic	Role	User Type	User Story	Acceptance Criteria	Module
US-22	EPIC-3	Company	Head of HR / Company rep	As a Company user, I want to approve, reject or delete recruiter accounts within our organization so that I control who acts on our behalf inside NeuroHire.	Pending-approvals queue lists recruiter accounts awaiting authorisation; approve grants dashboard access; disable sets `status = inactive` without deleting data; delete permanently removes the account after a confirmation prompt; all actions logged.	Company Console
US-23	EPIC-3	Company	Head of HR / Company rep	As a Company user, I want to view all job postings created by recruiters in our organization so that I have full oversight of current recruitment activity.	Jobs dashboard lists all active, inactive, and deleted jobs; filterable by status and recruiter; read-only for Company users.	Company Console
US-24	EPIC-3	Company	Head of HR / Company rep	As a Company user, I want an aggregated overview of applicant activity across all our job postings so that I can monitor overall recruitment pipeline health.	Overview aggregates total applications per job, shortlist counts; data refreshes at least every 5 minutes.	Analytics & Reporting
US-25	EPIC-3	Company	Head of HR / Company rep	As a Company user, I want to report a platform issue to NeuroHire so that it can be investigated and resolved by the Super Admin.	"Report Issue" form captures description and optional attachment; creates an `Error_Log` entry tagged with `company_id`; Company user can track issue status (open / in progress / resolved).	Error Reporting
US-26	EPIC-3	Company	Head of HR / Company rep	As a Company user, I want to update our company profile (name, logo, contact details) so that our branding remains current across the platform.	Profile edit form validates required fields; logo upload enforces size and format constraints; changes reflected in booth branding within one sync cycle; update logged.	Company Console
EPIC-4 Super Admin (NeuroHire Platform)						
US-27	EPIC-4	Super Admin	Super Admin	As a Super Admin, I want a global platform overview showing all registered companies, active recruiters, and system health so that I can monitor platform-wide usage.	Dashboard displays total companies (active / pending / suspended), total recruiters, total applications processed (daily/monthly), and system uptime; data exportable as CSV.	Super Admin Console
US-28	EPIC-4	Super Admin	Super Admin	As a Super Admin, I want to view error logs reported by recruiters and Company users in separate views so that I can triage issues by their source efficiently.	Two distinct log queues: Recruiter Error Log and Company Error Log; each shows reporter name, company, description, timestamp, and status; both support filtering by date and status.	Error Logs
US-29	EPIC-4	Super Admin	Super Admin	As a Super Admin, I want to update the resolution status of any reported error so that reporters are kept informed and the issue backlog is managed.	Status toggle updates the `Error_Log` record to Resolved or Unresolved; resolution notes can be added; the reporting user is notified by email on status change.	Error Logs
US-30	EPIC-4	Super Admin	Super Admin	As a Super Admin, I want to approve, reject, or delete company accounts so that the platform only hosts legitimate and active organizations.	Approve sets `company.status = active`; reject sets status to `rejected` with a stated reason and notifies the company; delete permanently removes the company and all associated recruiter accounts after a confirmation prompt; all actions logged.	Super Admin Console
US-31	EPIC-4	Super Admin	Super Admin	As a Super Admin, I want to configure global platform settings (e.g. default AI thresholds, supported languages) so that system-wide behavior can be adjusted without a code deployment.	Settings console exposes key-value pairs in `System_Settings`; changes validated before saving; change history maintained; settings take effect on the next processing cycle.	System Settings

3.2 Non-functional Requirements

Requirements are grouped by category and are grounded in the constraints and objectives of the project.

- **Usability:**
  - **NFR-U1 – Bilingual Interface:** The candidate pod interface shall support both Urdu and English, with all on-screen text, prompts, and audio guidance rendered in the language selected by the candidate at session start. Switching language shall take effect immediately without restarting the session.
  - **NFR-U2 – Low-Literacy Accessibility:** The candidate pod interface shall be navigable by users with limited literacy through the combined use of audio guidance, icons, and minimal written text. No candidate shall be required to read extended passages to complete an application.
  - **NFR-U3 – Guided Application Flow:** The candidate interface shall present a linear, step-by-step workflow that prevents candidates from skipping mandatory stages. Progress indicators shall be visible at all times during the application.
  - **NFR-U4 – Recruiter Dashboard Clarity:** The recruiter dashboard shall present candidate rankings, AI scores, and action controls in a single unified view, enabling a recruiter to review and action an application without navigating to more than two additional screens.
- **Security:**
  - **NFR-S1 – Role-Based Access Control:** The system shall enforce role-based access control across all interfaces. Candidates, Recruiters, Company/Head of HR, and Super Admin roles shall each have access only to the features and data permitted for their role. Unauthorized access attempts shall be rejected and logged.
  - **NFR-S2 – Secure Data Transmission:** All data transmitted between the candidate pod, recruiter dashboard, admin console, and the backend server shall be encrypted using HTTPS/TLS. Plain-text transmission of candidate data shall not be permitted.
  - **NFR-S3 – Candidate Data Privacy:** Candidate CVs, video recordings, and AI-generated scores shall be accessible only to authenticated recruiters and admins within the scope of their role.
  - **NFR-S4 – Authentication:** Recruiter and admin accounts shall require password-based authentication before granting access to the dashboard or console. Passwords shall be stored as cryptographic hashes; plain-text storage is prohibited.
- **Performance:**
  - **NFR-P1 – AI Processing Turnaround:** Following the submission of a candidate's CV and video responses, the system shall complete the full AI assessment pipeline — including OCR, speech-to-text transcription, and

behavioral scoring — and make results available on the recruiter dashboard within a targeted time of under five minutes per candidate under normal operating conditions.

  - **NFR-P2 – Pod Responsiveness:** The candidate pod interface shall respond to user interactions within two seconds under normal network conditions. Loading times for job listings and guidance screens shall not exceed three seconds.
  - **NFR-P3 – Video Upload Reliability:** The system shall handle video file uploads reliably, providing retry mechanisms in the event of transient network failure. A partially uploaded file shall not be accepted as a complete submission.

- **Reliability:**
  - **NFR-R1 – Error Logging and Auditability:** The system shall log all processing errors. Logs shall be retained and accessible to the Super Admin for auditing and issue resolution purposes.
  - **NFR-R2 – Graceful Degradation:** In the event of an AI processing failure (e.g., unintelligible audio, corrupted video file), the system shall not discard the application. Instead, it shall tag the application for manual HR review and surface it on the recruiter dashboard with an appropriate status indicator.
  - **NFR-R3 – Data Integrity:** Each submitted application shall be atomically linked to its associated CV data, video submission, AI assessment, and recruiter decision. Partial or orphaned records shall not be created during normal operation or recoverable failure states.

- **Maintainability:**
  - **NFR-M1 – Modular AI Pipeline:** The AI processing components — CV parsing, speech-to-text, and behavioral analysis — shall be implemented as independently deployable modules with well-defined interfaces, so that any single module can be updated or replaced without requiring changes to the others.

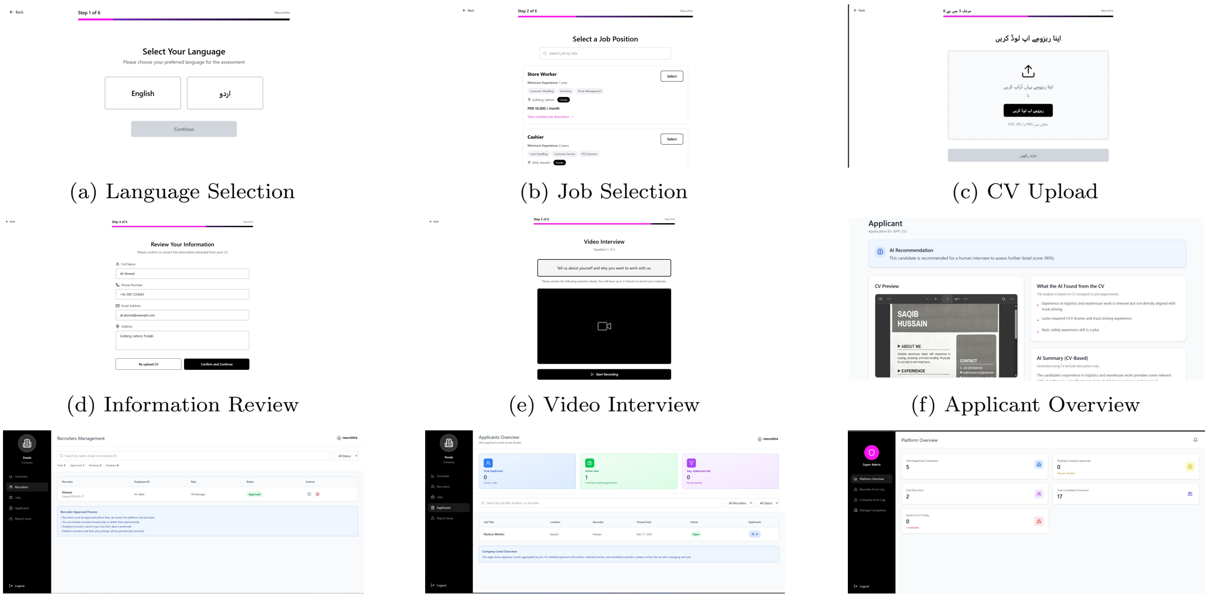
- **Fairness and Inclusivity:**
  - **NFR-F1 – Language Parity:** The system shall apply equivalent AI scoring criteria to Urdu-language and English-language submissions. Systematic score differentials attributable solely to language choice shall be identified and mitigated during model evaluation.
  - **NFR-F2 – Standardised Candidate Experience:** All candidates accessing the system through a recruitment pod shall be presented with the same hardware environment, interface layout, and sequence of prompts, ensuring that no candidate is advantaged or disadvantaged by the evaluation setup itself.
  - **NFR-F3 – Human Override Availability:** The system shall always preserve the recruiter's ability to override an AI recommendation. No AI decision shall be treated as final without the possibility of human review and intervention.

3.3 External Interfaces

3.3.1 User Interfaces

This section presents key user interface screens demonstrating the candidate workflow and recruiter/admin dashboards.

You can view the full design here: https://fetch-detach-39392601.figma.site/



(a) Language Selection (b) Job Selection (c) CV Upload

(d) Information Review (e) Video Interview (f) Applicant Overview

(g) Company Dashboard (h) Recruiter Management (i) Admin Panel

Figure 3.1: User interface screens of the NeuroHire system.

3.3.2 Hardware/Communication Interfaces

Since NeuroHire is designed to support video-based interview assessment, the recruitment pods must include a set of essential input, output, and networking components to capture and transmit candidate data reliably.

- **Camera and Microphone:** These components are required to record the candidate's video responses and clear audio during the interview session. The captured video supports facial and behavioral analysis, while the audio input is used for speech-to-text conversion and answer evaluation.
- **Screen and Speaker:** These components are used to display interview instructions, questions, and other interface elements to the candidate. The speaker provides audio guidance and feedback, which is especially useful in assisted or multilingual interview settings.
- **Internet Access:** A stable network connection is required to enable seamless communication between the recruitment pods and the cloud-based backend system. This allows uploaded CVs, recorded videos, transcripts, and analysis results to be transferred efficiently for storage and processing.

3.4 Use Case Diagrams

This section presents the use case diagrams for different user roles in the NeuroHire system.

3.4.1 Candidate Use Case

The candidate use case diagram shows the primary interactions of a candidate with the system. The candidate can select a preferred language (English or Urdu) and enable voice guidance for accessibility. The candidate can browse job listings, upload a CV, and review or edit extracted CV information before proceeding. After submission, the candidate records a video interview by answering screening questions, uploads responses, and receives confirmation. The diagram also highlights that CV upload includes information review, while video interviews include answering questions and submission.

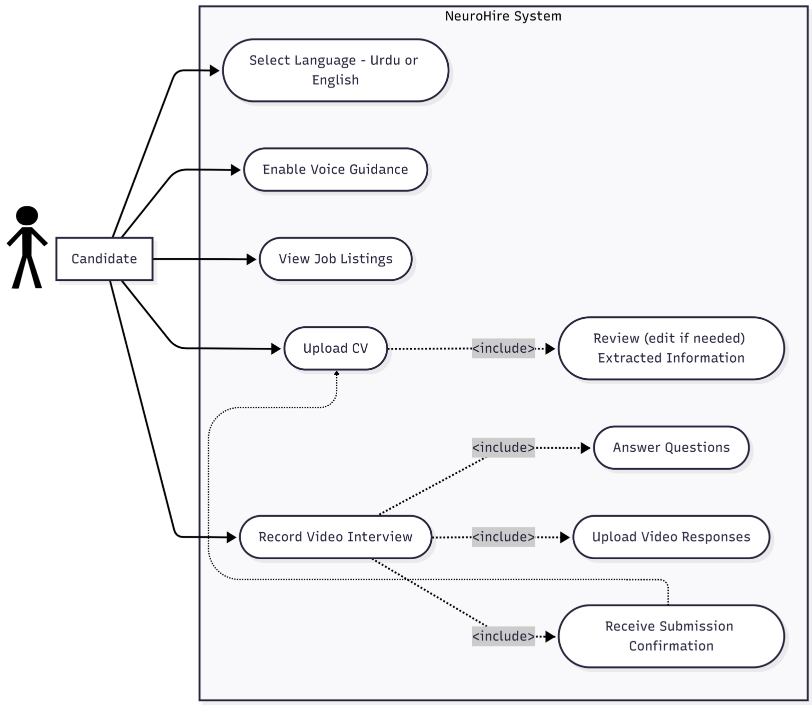


Figure 3.2: Candidate Use Case Diagram

3.4.2 Recruiter Use Case

The recruiter use case diagram represents recruiter actions within the system. The recruiter can log in or sign up, post job vacancies, view applicants, make hiring decisions, and report system errors. Posting a job includes adding job descriptions, screening questions, and scoring thresholds, and can be extended to editing, enabling/disabling, or deleting jobs. While reviewing applicants, the recruiter can access CVs, video interviews, AI recommendations, summaries, and confidence scores. Based on this information, the recruiter can accept, reject, or forward candidates for further interviews.

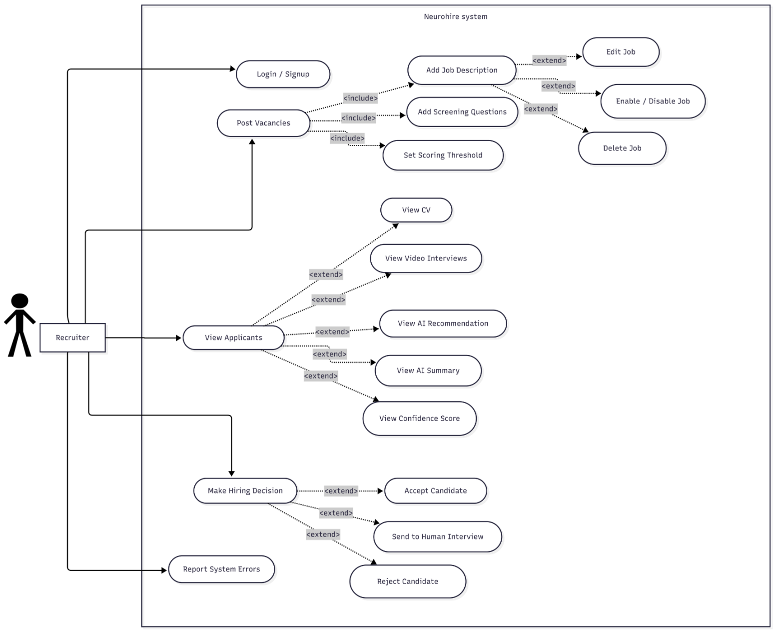


Figure 3.3: Recruiter Use Case Diagram

3.4.3 Admin Use Case

The admin use case diagram illustrates company-level administrative functions. The admin can manage recruiters, view jobs, access applicant overviews, and report issues. Recruiter management can be extended to approving, disabling, or deleting recruiters, allowing control over recruiter access and monitoring of hiring activity.

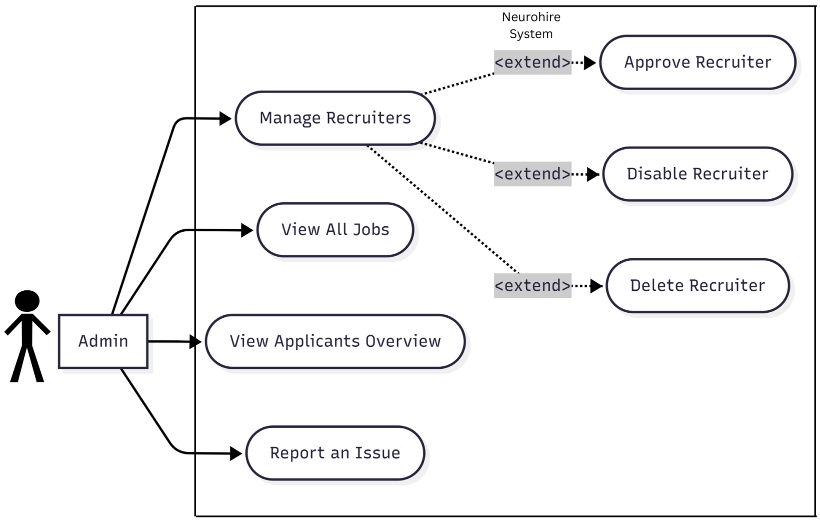


Figure 3.4: Admin Use Case Diagram

3.4.4 Super Admin Use Case

The super admin use case diagram represents platform-level control. The super admin can view the system dashboard, access recruiter and company error logs, and manage companies. Error logs can be updated by marking issues as resolved or unresolved. Company management includes approving, rejecting, or deleting companies, enabling full platform monitoring and control.

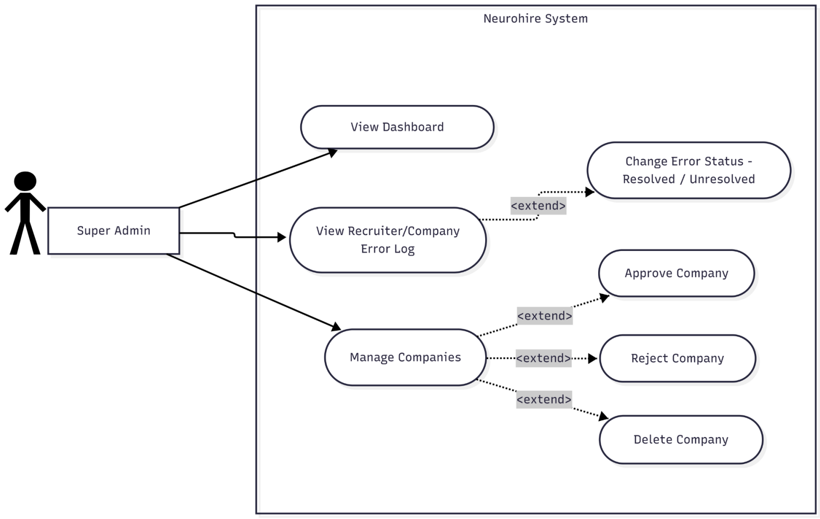


Figure 3.5: Super Admin Use Case Diagram

3.5 System Diagram

This diagram gives a high-level view of the different components of our system and the interactions between them.

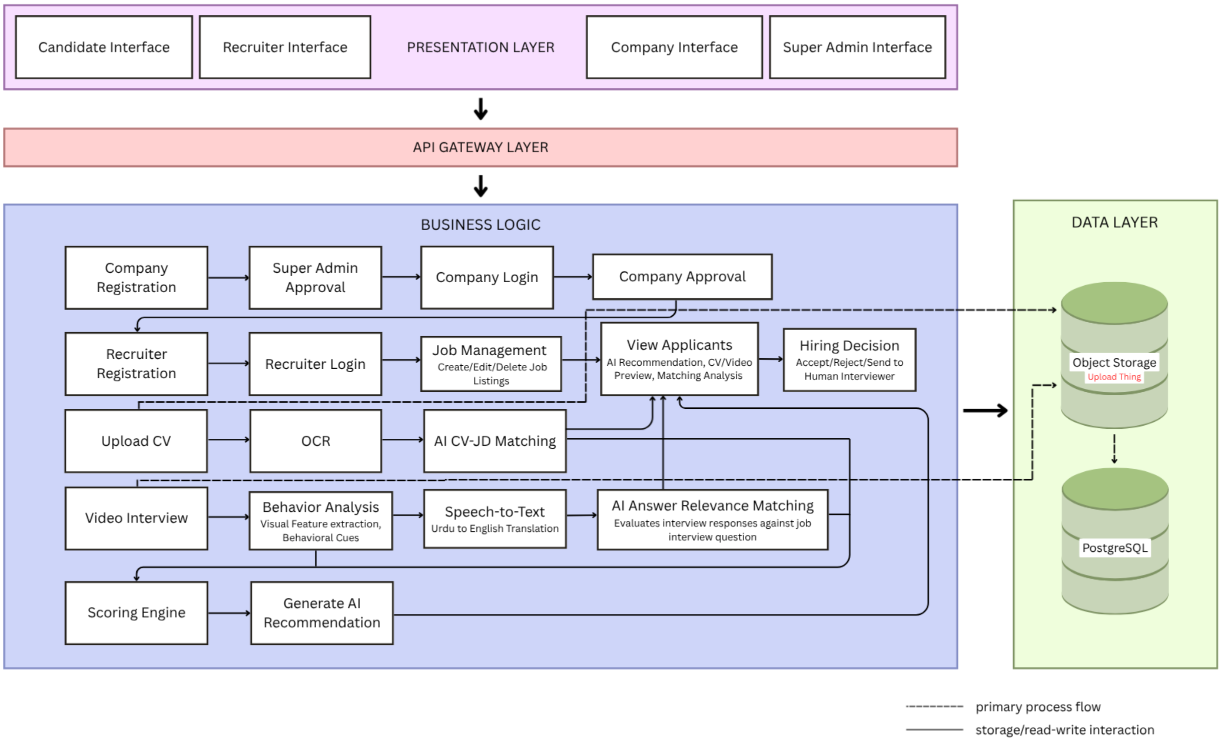


Figure 3.6: High-level system architecture of NeuroHire

Each component and the particular tools, technologies, or libraries used to build it are described below:

- **OCR Module:** This component is responsible for extracting textual content from uploaded CVs and resume documents. It is implemented using **OCRmyPDF** and **PaddleOCR** for optical character recognition, while **PyMuPDF** is used for PDF text extraction and document preprocessing.

- **CV–JD Matching Module:** This component compares the candidate's CV with the job description to evaluate relevance and suitability. It is implemented using an **LLM-Based CV–JD Matching** approach built on **Fine-tuned LLaMA 3.2 – 7B**.

- **Speech-to-Text Module:** This component converts spoken interview responses into text for further analysis. It is implemented using **OpenAI Whisper Large v3**, which supports multilingual transcription.

- **Answer Relevance Matching Module:** This component evaluates transcribed interview responses against job interview questions to assess answer relevance and response quality. It is implemented using **Phi-3.5 Mini Instruct (Q4_K_M GGUF)**.

- **Behavior Analysis Module:** This component extracts visual and behavioral cues from interview videos, such as gaze direction, head movement, and facial stability. It is implemented using **OpenCV** and **MediaPipe**.

- **Database Layer:** This component stores structured system data such as user accounts, job listings, applicant records, scores, and hiring decisions. It is implemented using **PostgreSQL**.
- **Object Storage Layer:** This component stores unstructured files such as uploaded CVs and recorded interview videos. It is implemented using **UploadThing**.

4. Software Design Specification (SDS)

This chapter provides important artifacts related to the design of our project.

4.1 Data Design

This section presents the structure of the database designed for NeuroHire, which supports persistent data storage and efficient retrieval of information across the system. The database is implemented using **PostgreSQL** and deployed on **Neon**, a managed cloud-based Postgres platform.

The data model is represented as a normalized Entity-Relationship Diagram (ERD), illustrating key entities, attributes, relationships, primary keys, and foreign key constraints. The design ensures data integrity, reduces redundancy, and supports recruitment workflows.

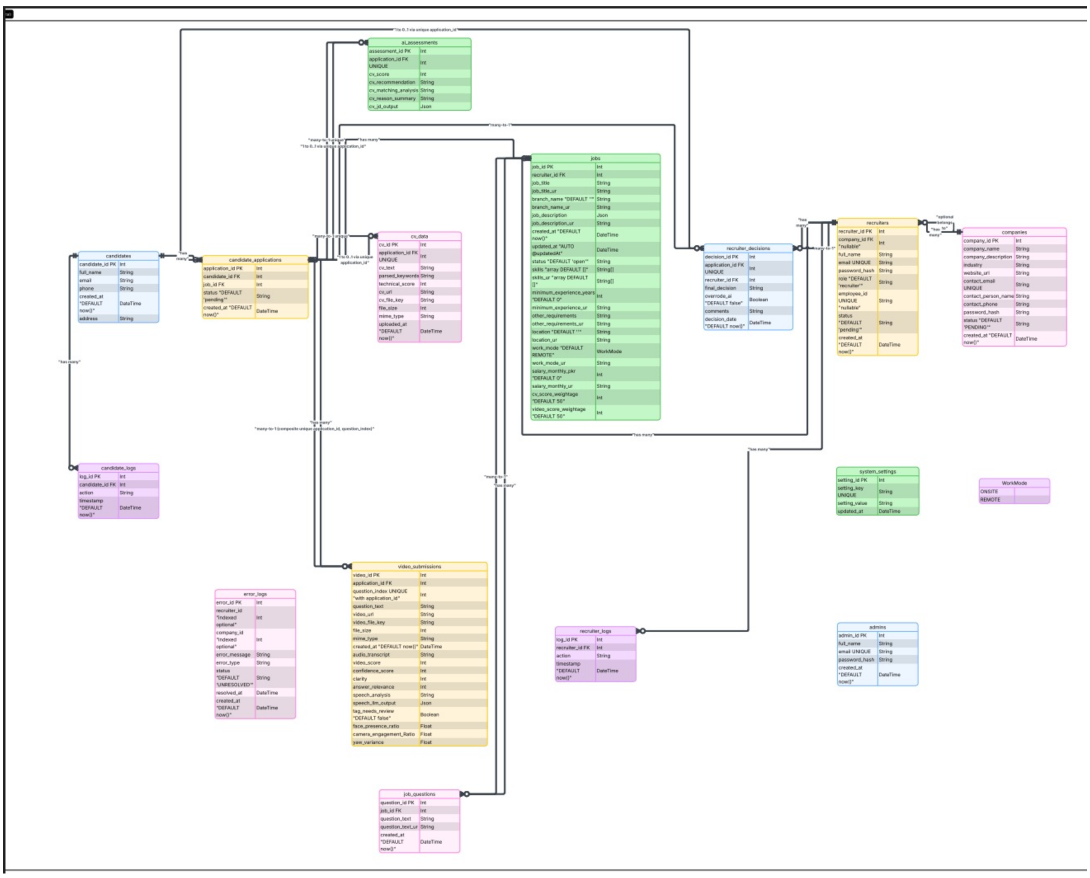


Figure 4.1: Entity-Relationship Diagram of NeuroHire Database

Key Entities and Relationships

The core entities in the system include:

- **Candidates**: Stores candidate personal information and profile details.
- **Jobs**: Contains job postings created by recruiters, including role requirements and metadata.
- **Candidate Applications**: Represents candidate applications for specific jobs and tracks application progress.
- **CV Data**: Stores parsed and structured information extracted from uploaded resumes.
- **Video Submissions**: Stores recorded interview responses and extracted video-related features.
- **Assessments**: Stores AI-generated evaluation results, including CV-job matching scores, video analysis, and recommendations.
- **Recruiters and Companies**: Represents the organizational hierarchy where recruiters belong to companies and manage job postings.
- **Recruiter Decisions**: Captures final hiring decisions made by recruiters based on AI-assisted insights.
- **Job Questions**: Stores structured interview questions associated with each job.

Design Highlights

- The schema follows **relational normalization principles** to reduce redundancy and maintain consistency.
- **Foreign key constraints** enforce relationships between candidates, jobs, applications, recruiters, and companies.
- The design supports **multimodal data storage**, including structured CV data, video metadata, and AI-generated insights.
- Logging tables such as candidate logs and error logs support **system monitoring and debugging**.
- The schema supports **AI-driven recruitment workflows**, including scoring, recommendation, and decision tracking.

4.2 Technical Details

This section describes the technical architecture and core processing pipelines used in NeuroHire, focusing on system design, data flow, and algorithmic components.

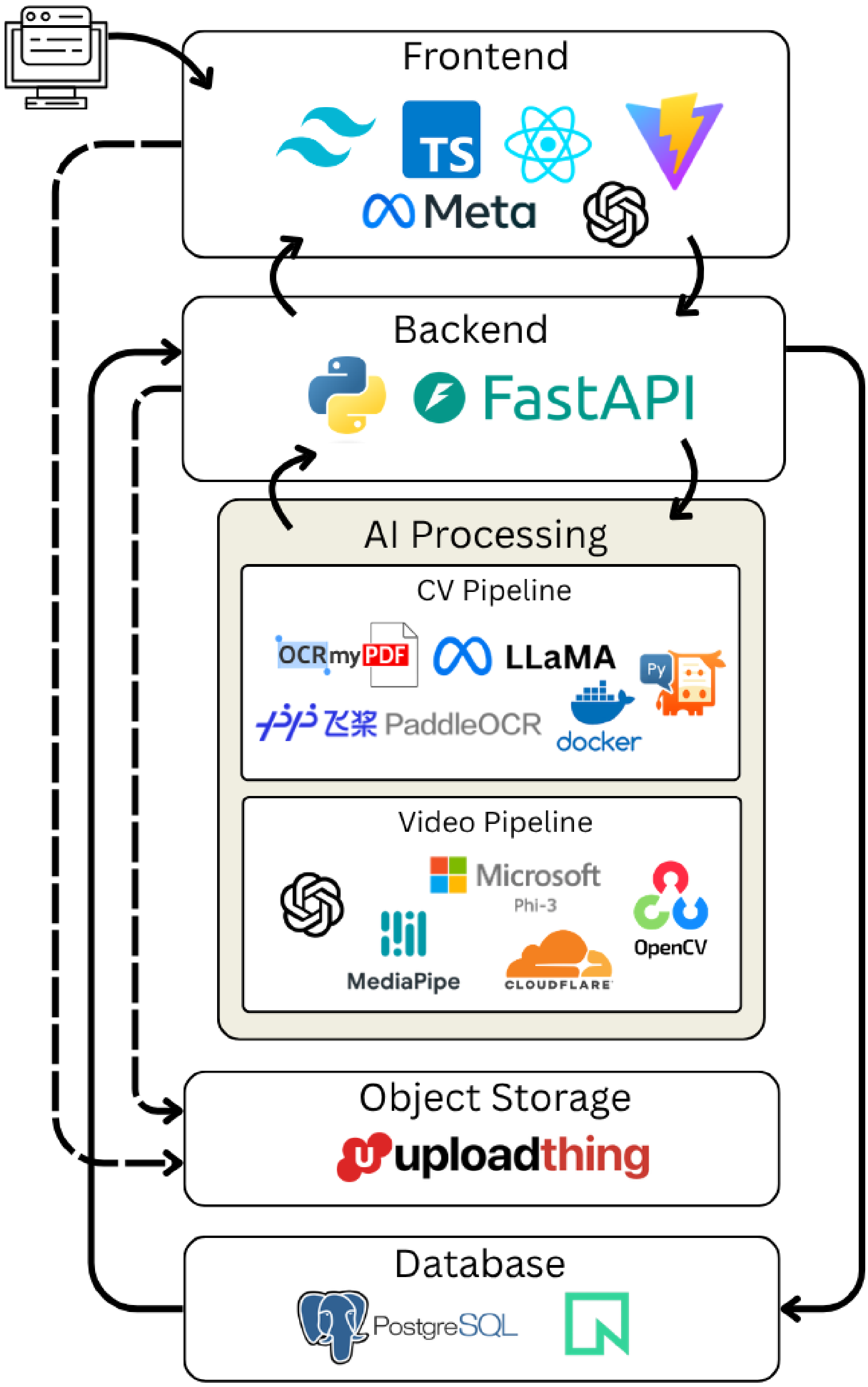


Figure 4.2: Overall Technical Architecture of NeuroHire

The system follows a layered architecture consisting of a frontend interface, backend API, AI processing pipelines, object storage, and database. The frontend communicates with the backend through API calls, while the backend coordinates AI pipelines for CV and video processing.

The AI pipelines are deployed on a **High Performance Computing (HPC) en-**

vironment with GPU acceleration. The **CV pipeline is accessed through a Gradio-based API**, while the **video pipeline is integrated through FastAPI and exposed using Cloudflare for secure routing**.

Object storage is used for storing CVs and video files, while PostgreSQL on Neon is used for structured data and results.

4.2.1 Module 1: CV Processing Pipeline

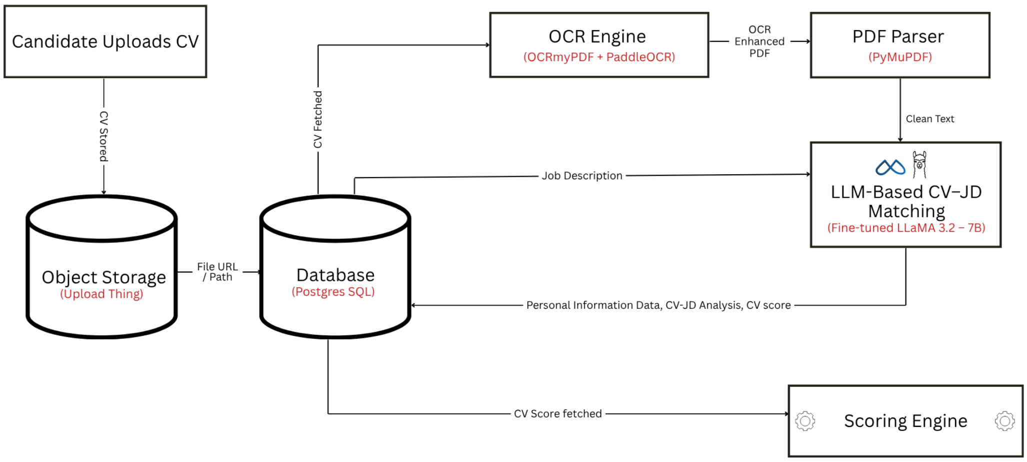


Figure 4.3: CV Processing Pipeline

The CV pipeline processes uploaded resumes and performs job matching using AI. Candidate CVs are stored in object storage and referenced in the database. OCR techniques such as OCRmyPDF and PaddleOCR are applied to enhance and extract text from scanned documents. PyMuPDF is then used to parse the resume content into a cleaner structured format.

The processed CV data, along with the job description, is passed to a fine-tuned LLaMA 3.2 model, which performs CV–Job Description matching. The model outputs a CV score and analysis, which are stored in the database and later used in final evaluation.

4.2.2 Module 2: Video Processing Pipeline

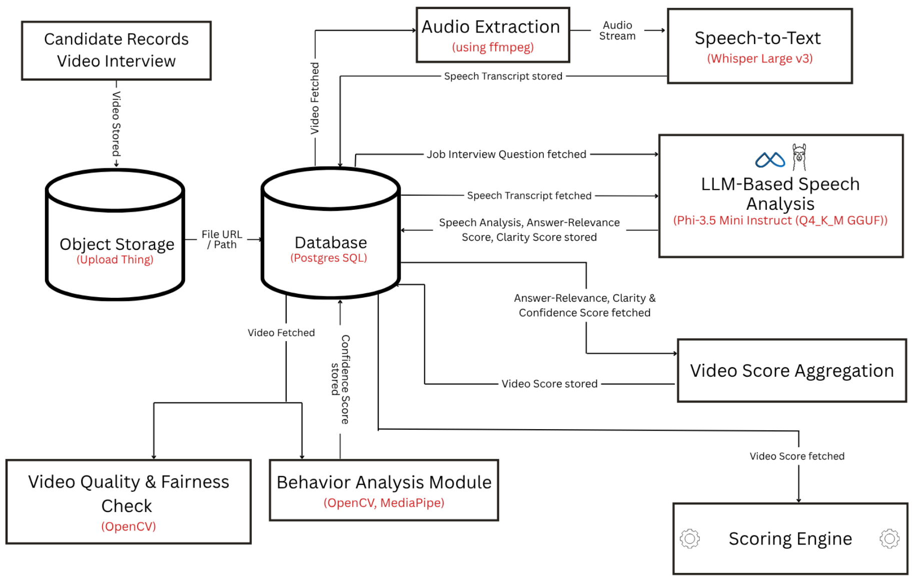


Figure 4.4: Video Processing Pipeline

The video pipeline evaluates candidate responses through multimodal analysis. Interview videos are stored in object storage and accessed through the database. Audio is extracted using FFmpeg and converted into text using Whisper for speech-to-text processing.

The transcript and interview questions are analyzed using the Phi-3.5 Mini model to compute answer relevance and clarity scores. In parallel, OpenCV and MediaPipe are used for behavioral analysis and video quality checks, extracting features such as confidence and facial cues.

All outputs are aggregated into a unified video score, which is stored in the database and forwarded to the scoring engine.

4.2.3 Module 3: Scoring Engine

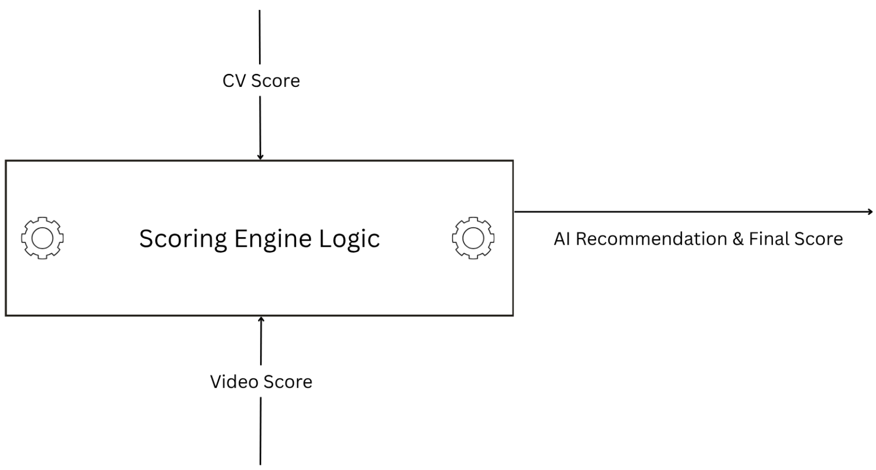


Figure 4.5: Scoring Engine Logic

The scoring engine integrates outputs from both the CV and video pipelines to generate a final candidate score and recommendation.

CV Score: The CV score is obtained directly from the CV–Job Description matching model, which returns a *Total_score*. This value is normalized and stored in the database without additional computation.

Video Score Per Question: Each video response is analyzed to extract *confidence*, *clarity*, and *answer relevance*. The per-question score is computed as the average of the available metrics.

Application-Level Video Score: The overall video score is calculated as the mean of all per-question scores across the interview.

Final Score Calculation: Each job defines weightages for CV and video scores, which sum to 100. The final score is computed as:

$$\text{Total Score} = \text{CV Score} \times w_{cv} + \text{Video Score} \times w_{video}$$

If the CV score is unavailable, the system falls back to using only the video score.

Recommendation Generation: The final score is mapped to recommendation bands such as shortlist, interview, or reject. This provides interpretable decision support while maintaining human oversight.

5. Methodology, Experiments and Results OR Testing and Analysis

5.1 CV-JD Matching Module

The CV-JD matching module was evaluated to measure how accurately the system could determine whether a candidate's CV was suitable for a given job description. This module is important because NeuroHire is designed for high-volume hiring, where recruiters may need to screen a large number of candidates against multiple no-skill or low-skill job roles. The purpose of this experiment was to compare the system's accept/reject recommendation with human-labeled ground truth decisions.

For this experiment, a dataset of 40 CVs was tested against 12 job descriptions, producing a total of 480 CV-JD combinations. Each CV-JD pair was assigned a score by the model and also manually evaluated using a human score. A threshold of 50 was used to convert both model and human scores into binary labels. Scores greater than or equal to 50 were treated as *Accept*, while scores below 50 were treated as *Reject*. The system output was then compared with the human label to calculate classification performance.

The model's performance was evaluated using accuracy, precision, recall, F1-score, specificity, confusion matrix, and ROC-AUC. Accuracy measures the overall number of correct predictions. Precision measures how many candidates predicted as acceptable were actually acceptable. Recall measures how many human-accepted candidates were successfully identified by the model. Specificity measures how well the model rejected unsuitable candidates. The ROC-AUC score was used to evaluate the model's ability to separate acceptable and rejectable candidates across different score thresholds.

5.1.1 Confusion Matrix Results

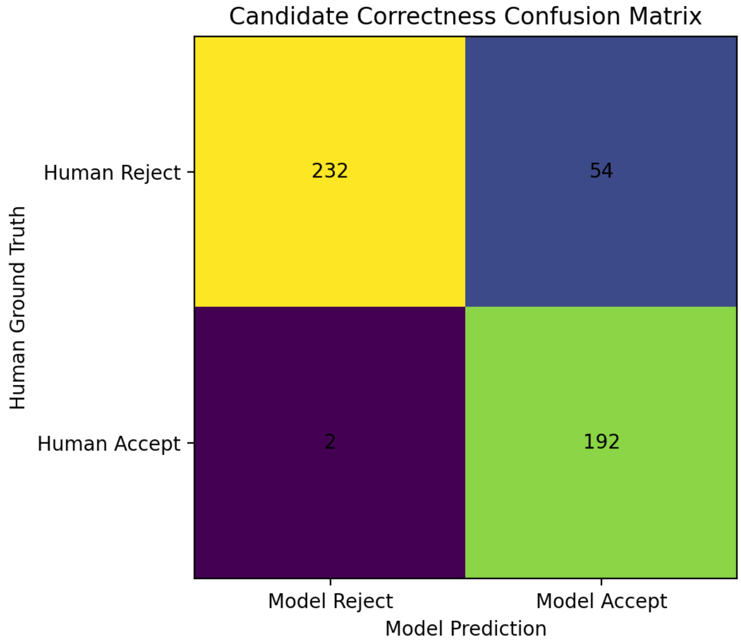


Figure 5.1: Confusion matrix for the CV-JD matching module comparing model predictions with human ground truth labels.

The confusion matrix shows that the model correctly rejected 232 unsuitable CV-JD pairs and correctly accepted 192 suitable pairs. There were 54 false positives, where the model accepted candidates that human evaluation rejected. There were only 2 false negatives, where the model rejected candidates that human evaluation accepted.

This result is useful for NeuroHire because, in a pre-screening system, missing a potentially suitable candidate can be more harmful than forwarding an extra candidate for recruiter review. The low number of false negatives shows that the model is strong at identifying candidates who should not be missed.

5.1.2 Performance Metrics

Table 5.1: Performance metrics for the CV-JD matching module

Metric	**Score**
Accuracy	0.883
Precision	0.780
Recall	0.990
F1-Score	0.873
Specificity	0.811
ROC-AUC	0.848

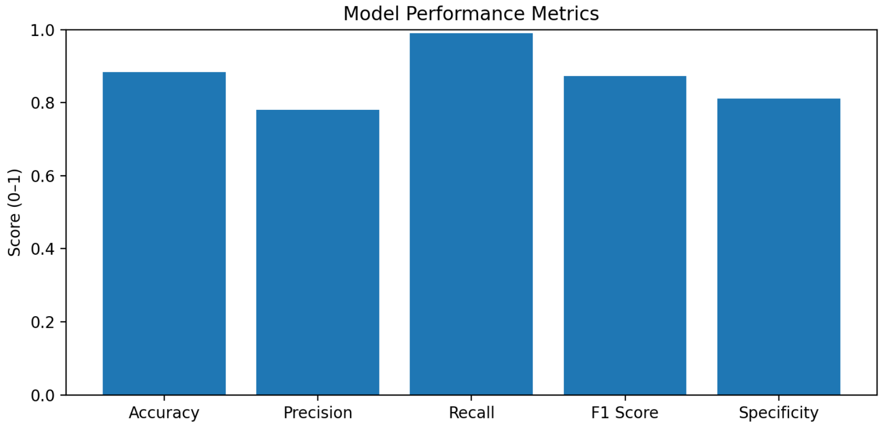


Figure 5.2: Performance metrics of the CV-JD matching module, including accuracy, precision, recall, F1-score, and specificity.

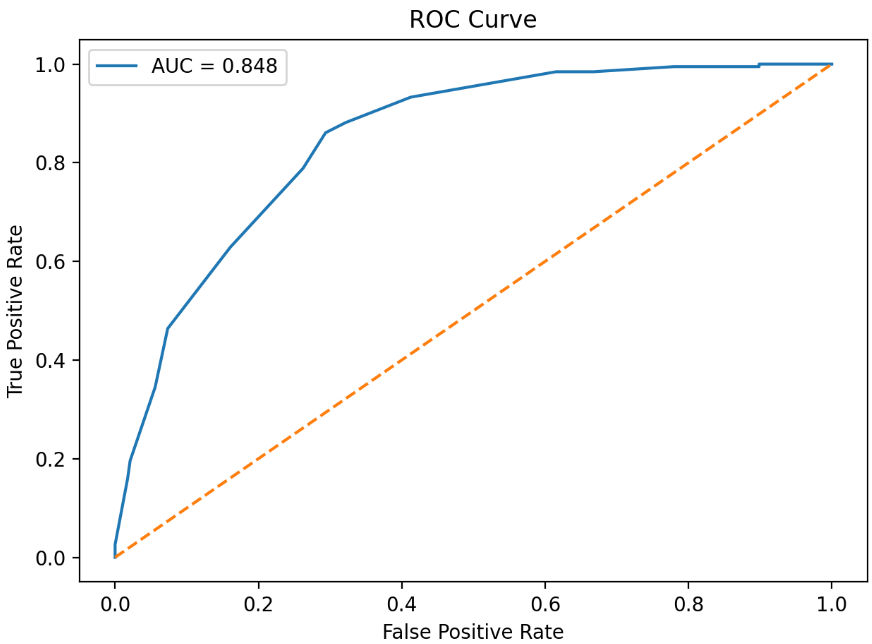


Figure 5.3: ROC curve for the CV-JD matching module showing an AUC score of 0.848.

The module achieved an overall accuracy of 88.3%, showing that the system's final accept/reject decisions matched human judgment in most cases. The recall score of 99.0% was the strongest result, meaning the model identified almost all candidates that humans considered acceptable. This is especially important in recruitment because the system should avoid filtering out suitable applicants too early.

The precision score was 78.0%, which means some candidates accepted by the model were not accepted by human evaluators. However, this is still acceptable for a shortlisting system because the final hiring decision remains with the recruiter. The specificity score of 81.1% shows that the model was also reasonably effective at rejecting unsuitable CV-JD pairs. The F1-score of 87.3% indicates a strong balance between precision and recall.

The ROC curve produced an AUC score of 0.848, which indicates that the model has good separation ability between acceptable and rejectable CV-JD pairs. Since the AUC is clearly above 0.5, the model performs significantly better than random classification.

5.1.3 Discussion

The results show that the CV-JD matching module is effective for initial candidate screening. Its strongest performance is in recall, meaning it is highly unlikely to miss candidates that human evaluators consider suitable. This aligns well with NeuroHire's goal of supporting recruiters rather than replacing them. The module can reduce manual workload by filtering and organizing candidates, while recruiters can still review borderline or incorrectly accepted cases.

The main limitation observed is the number of false positives. The model accepted 54 candidates that human evaluators rejected. This suggests that the system may sometimes give importance to partial keyword or experience matches even when the overall candidate-job fit is weak. In a real recruitment setting, this can be handled by showing these candidates to recruiters with their CV-JD score and explanation rather than automatically making a final decision.

Overall, the CV-JD module demonstrates strong potential as a pre-screening component in NeuroHire. It performs well in identifying suitable candidates, provides a measurable scoring mechanism, and supports the platform's broader goal of making high-volume hiring faster, more consistent, and more transparent.

5.2 Video Analysis Module

The video analysis module evaluates candidate video responses using three components: confidence, clarity, and answer relevance. Confidence is evaluated through internal behavioral and visual metrics extracted from the video, while clarity and answer relevance are evaluated through the transcribed response. Clarity measures whether the answer is understandable and well-structured, while answer relevance measures whether the answer directly addresses the interview question.

5.2.1 Confidence Evaluation

The confidence component evaluates visual and behavioral cues from the candidate's video response. Since confidence is subjective and can be influenced by factors such as nervousness, posture, camera angle, and presentation style, this component was validated through recruiter-based testing. Recruiters from KFC tested the NeuroHire product and reviewed whether the system-generated confidence score aligned with their own judgment of candidate confidence.

5.2.2 Whisper Transcription and Translation Evaluation

The Whisper component was evaluated to measure the reliability of the speech processing stage in NeuroHire's video analysis pipeline. This component is important because later stages of video analysis, especially clarity and answer relevance, depend on the text generated from the candidate's spoken response. If the transcription or translation output is incomplete or inaccurate, the downstream answer evaluation may also become unreliable.

The evaluation was divided into two separate parts. English audio samples were used to test Whisper's transcription performance, where the system was expected to convert spoken English into written English. Urdu audio samples were used mainly to test Whisper's translation performance, where the system was expected to process Urdu speech and produce usable English text for downstream analysis. This separation was necessary because NeuroHire supports both English and Urdu-speaking candidates, while the later answer evaluation module works on text responses in a common format.

For this experiment, 400 English audio samples were selected from the Mozilla Common Voice dataset to evaluate English transcription. Similarly, 400 Urdu audio samples were selected from an Urdu speech dataset to evaluate Urdu-to-English translation. In total, 800 audio samples were used across both evaluations.

The output was evaluated as a binary quality classification task. For English transcription, an output was marked as *Acceptable* if the transcript preserved the main spoken content and was usable for answer evaluation. For Urdu translation, an output was marked as *Acceptable* if the translated English text preserved the main meaning of the Urdu speech. Outputs were marked as *Unacceptable* if they contained major transcription errors, missing content, or meaning-changing translation mistakes.

English Transcription Evaluation

The English transcription evaluation tested whether Whisper could convert English speech into usable written text. The goal was not perfect word-by-word transcription, but to determine whether the generated transcript preserved enough meaning for the clarity and answer relevance modules.

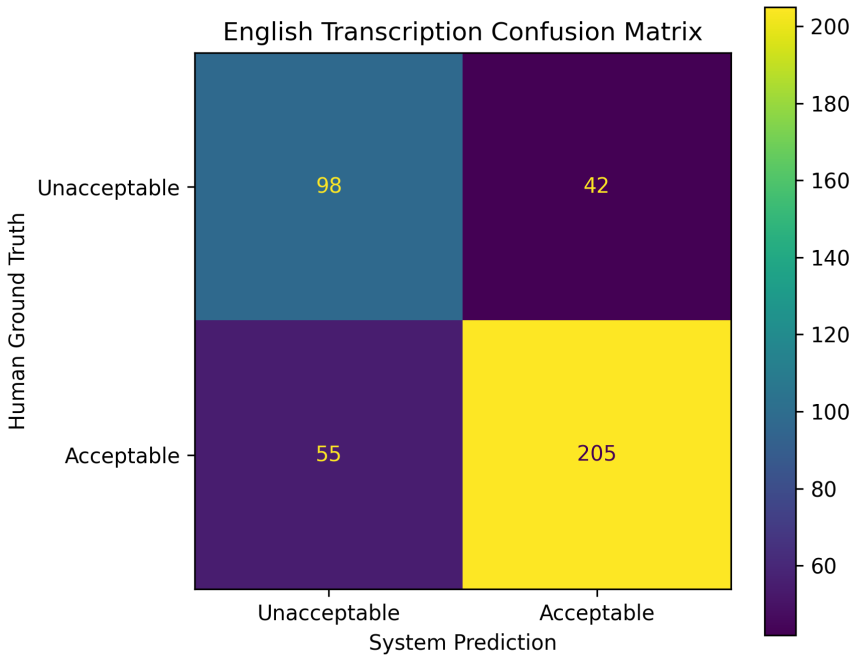


Figure 5.4: Confusion matrix for English transcription evaluation.

The English transcription confusion matrix shows the system's performance in classifying generated transcripts as acceptable or unacceptable. The module correctly identified most usable transcripts, but some valid transcripts were still marked as unacceptable due to minor wording differences or incomplete phrasing. A smaller number of poor transcripts were also marked as acceptable, which shows the need for transcript quality checks before passing outputs to the answer evaluation stage.

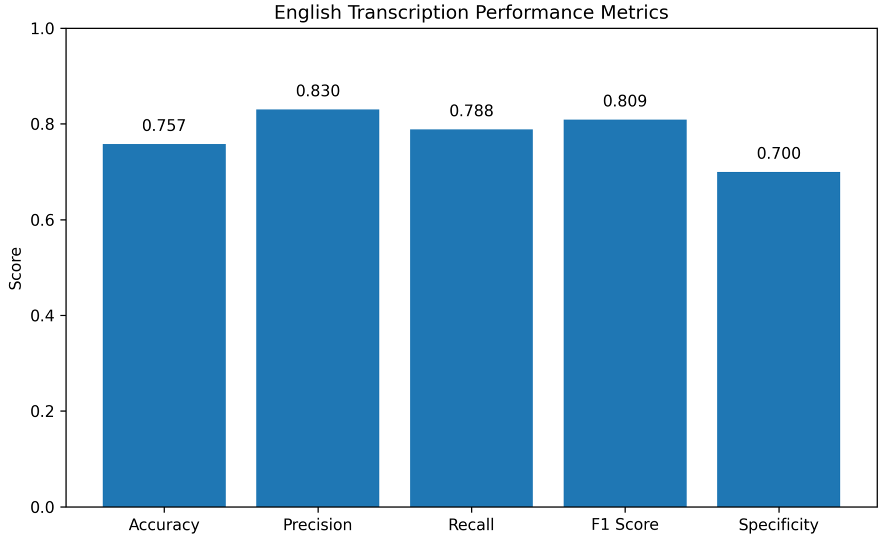


Figure 5.5: Performance metrics for English transcription evaluation.

The English transcription evaluation achieved an accuracy of 75.8%. The precision score of 82.6% shows that when the system marked an English transcript as acceptable, it was usually usable for downstream analysis. The recall score of 78.8% shows that most usable English transcripts were correctly identified. The F1-score of 80.6% indicates a reasonable balance between precision and recall.

Urdu-to-English Translation Evaluation

The Urdu evaluation focused mainly on translation rather than direct Urdu transcription. Since NeuroHire's downstream answer relevance module works on text responses in a common format, Urdu candidate responses were translated into English after speech processing. The purpose of this test was to check whether Whisper could preserve the main meaning of Urdu speech when converting it into English text.

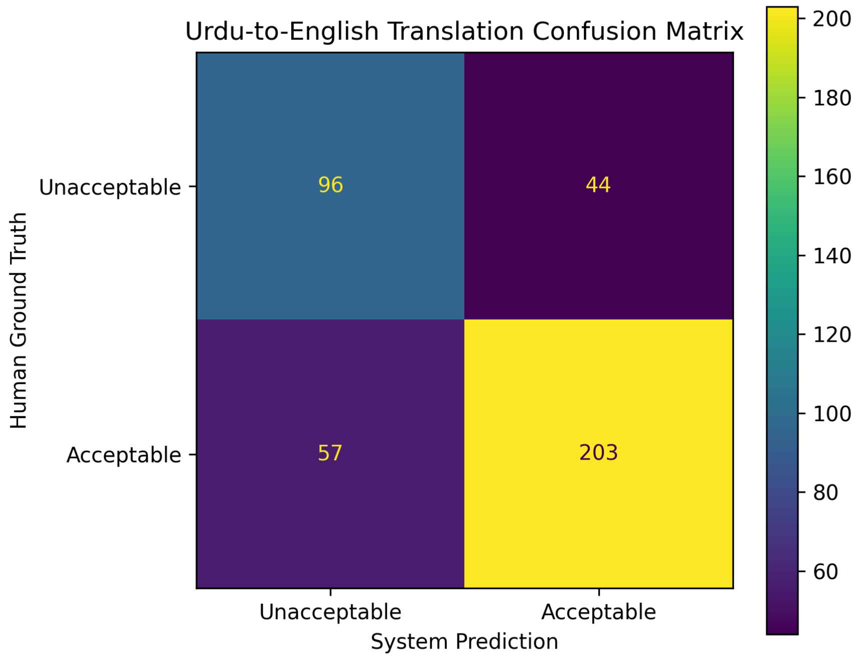


Figure 5.6: Confusion matrix for Urdu-to-English translation evaluation.

The Urdu-to-English translation confusion matrix shows that the module was able to correctly identify many usable translations, but it also produced more errors than the English transcription evaluation. This was expected because the task required the system to first understand Urdu speech and then produce an English translation that preserved the original meaning.

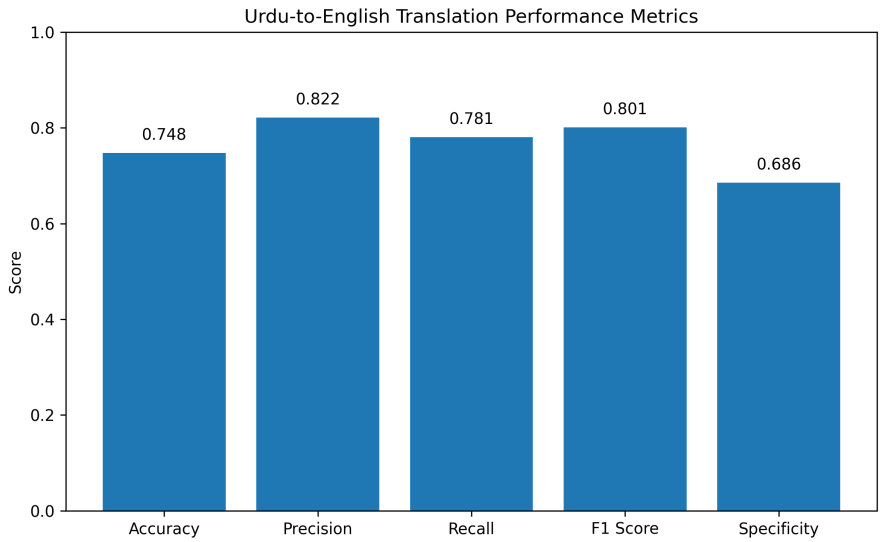


Figure 5.7: Performance metrics for Urdu-to-English translation evaluation.

The Urdu-to-English translation evaluation achieved an accuracy of 74.8%, which was slightly lower than the English transcription result. This difference is reasonable because translation is more complex than direct transcription. The precision score of 82.2% shows that outputs marked as acceptable were generally reliable, while the recall score of 78.0% shows that most meaning-preserving translations were correctly identified. The F1-score of 80.0% shows that the translation module maintained a useful balance between correctly accepting usable translations and rejecting poor outputs.

5.2.3 Answer Relevance Evaluation

The answer relevance module was evaluated as a sub-component of the video analysis pipeline. After the candidate's video response is transcribed, the system analyzes whether the answer is relevant to the interview question. This step is important because a candidate may speak clearly, but the response may still be off-topic, vague, or incomplete. Therefore, the purpose of this experiment was to test whether the system could distinguish between relevant and irrelevant interview answers.

Interview Question Dataset

For the answer relevance evaluation, a publicly available HR interview question dataset from Kaggle was used as the base source for interview prompts. The dataset provided a diverse set of behavioral and role-related interview questions, which were used to construct positive and negative response samples for testing the answer relevance module. The dataset was used only for generating and structuring interview-question scenarios, while the evaluation focused on whether the candidate response was relevant to the asked question. We chose random positive 150 and 100 negative samples from the dataset to create a balanced test set of 250 responses. Each response was manually labeled as relevant (positive) or irrelevant (negative) based on whether it appropriately addressed the interview question.

Confusion Matrix Results

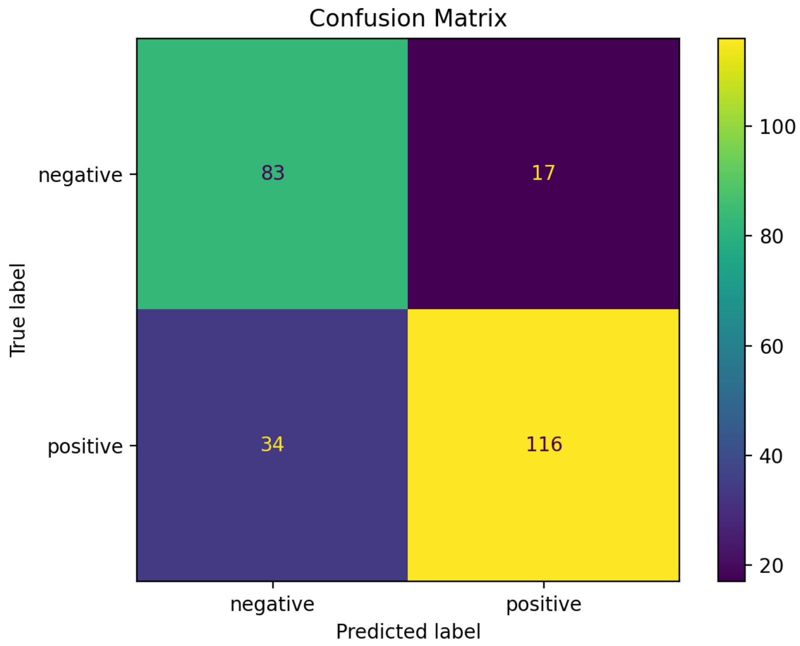


Figure 5.8: Confusion matrix for the answer relevance module comparing predicted labels with ground truth labels.

The confusion matrix shows that the module correctly identified 83 negative responses and 116 positive responses. There were 17 false positives, where weak or irrelevant answers were classified as positive. There were also 34 false negatives, where relevant answers were classified as negative.

The false positives are important because they represent cases where the system may allow a weak answer to pass. However, the number of false positives is relatively low compared to the number of correctly identified positive answers. The false negatives show that the system was sometimes stricter when evaluating relevant answers, especially when the answer was generally correct but lacked enough detail or depth.

Performance Metrics

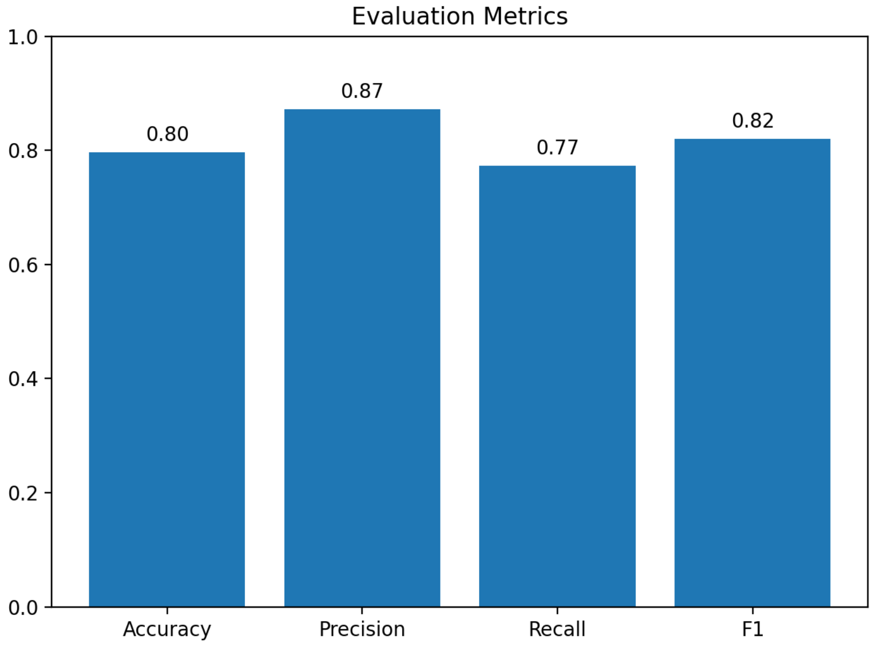


Figure 5.9: Evaluation metrics for the answer relevance module, including accuracy, precision, recall, and F1-score.

The answer relevance module achieved an overall accuracy of 80%, showing that the system correctly classified most interview responses. The precision score of 87% indicates that when the system classified an answer as relevant, it was usually correct. This is useful in recruitment because it reduces the chance of poor or irrelevant answers being treated as strong responses.

The recall score was 77%, which means the system identified most relevant answers but missed some positive cases. This suggests that the module is slightly conservative and may mark some acceptable answers as negative if they do not contain enough detail. The F1-score of 82% shows a reasonable balance between precision and recall.

KFC Recruitment Drive Video Evaluation

In addition to the labeled interview response dataset, the answer relevance module was also evaluated using real video data collected during a KFC recruitment drive. Candidate responses were recorded during the recruitment activity and later converted into question-answer pairs using the generated transcripts. These question-answer pairs were used to test whether the answer relevance module could perform consistently on real candidate responses, rather than only on prepared or structured test samples.

The purpose of this additional evaluation was to observe the module's behavior in a more realistic recruitment environment, where responses may include hesitation, informal phrasing, incomplete answers, background noise, or less structured communication. Since this data came from an actual recruitment setting, it provided a more practical validation of how the answer relevance module would perform in NeuroHire's intended use case.

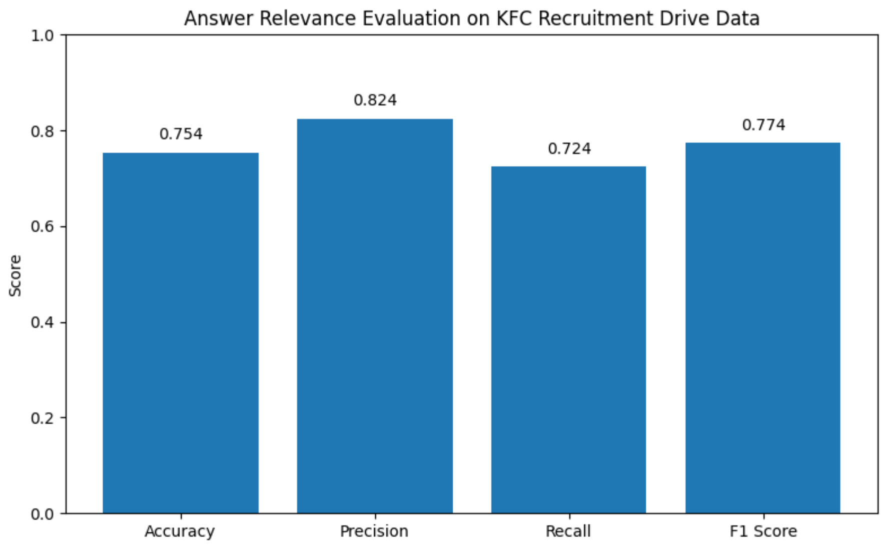


Figure 5.10: Evaluation metrics for the answer relevance module on KFC recruitment drive video data.

The KFC recruitment drive evaluation showed slightly lower performance compared to the structured dataset evaluation. This was expected because real candidate responses were less controlled and included variations in speaking style, answer length, and clarity. However, the module still maintained useful performance, showing that it can support recruiter decision-making in practical recruitment settings.

5.2.4 Discussion

Overall, the video analysis module shows that NeuroHire can support recruiters by evaluating candidate responses through confidence, transcription quality, translation quality, clarity, and answer relevance. The Whisper evaluation showed that English transcription performed slightly better than Urdu-to-English translation, which was expected because translation requires both speech recognition and meaning preservation. Since answer relevance depends on the generated transcript, these results show why transcript quality must be treated as an important part of the video pipeline.

The answer relevance module performed well on the structured dataset, achieving 80% accuracy, 87% precision, 77% recall, and an F1-score of 82%. Its high precision shows that when the system marks an answer as relevant, the judgment is usually reliable. The KFC recruitment drive evaluation showed a slight drop in performance, which was expected because real candidate responses included hesitation, informal wording, background noise, and shorter answers. However, the module still remained useful for practical recruitment support.

The confidence component was validated through recruiter-based testing with KFC recruiters. Since confidence is subjective and can be affected by nervousness, lighting, posture, camera angle, and comfort with video recording, it should not be used as an automatic rejection factor. Instead, it works better as a supporting indicator alongside transcription quality, clarity, and answer relevance.

Overall, the module should be treated as a decision-support tool rather than a replacement for recruiter judgment. It helps make high-volume hiring more structured and consistent, but borderline cases, unclear transcripts, low-confidence responses, or weak answer relevance scores should be forwarded for manual recruiter review to avoid unfairly filtering out potentially suitable candidates.

5.3 Application Testing

Application testing was conducted to verify whether the user-facing NeuroHire platform works correctly across its main roles: Candidate, Recruiter, Company, and Super Admin. While module-level testing evaluates the accuracy of individual AI components, application testing focuses on whether users can complete the intended workflows through the interface without functional errors.

The application was tested using both manual and automated testing. Manual testing was used to validate complete user flows from the perspective of real users, such as a candidate applying for a job or a recruiter reviewing an applicant. Automated testing was used to repeatedly check common interface actions such as login, navigation, form submission, job creation, dashboard loading, access control, and confirmation messages. Selenium was used for browser-based automated testing because it allows the system to simulate real user actions on the web interface and verify whether the expected page elements or responses appear.

It is important to note that Selenium was used for application workflow testing, not for evaluating AI model accuracy. The quality of AI outputs was evaluated separately through the module-level experiments discussed earlier. In this section, automated testing refers only to interface behavior, navigation, forms, user actions, and expected application responses.

The test plan below summarizes the main application-level scenarios tested during development and validation. These scenarios were tested manually, and selected repeated workflows were also checked through automated browser-based testing.

Table 5.2: Application testing plan for NeuroHire

Test ID	**Application Area**	**Test Scenario**	**Expected Result**	**Status**
AT-01	Candidate Interface	Candidate selects preferred language at the start of the session.	The interface updates according to the selected language and the candidate can continue the application flow.	Pass
AT-02	Candidate Interface	Candidate views available job listings.	Active jobs are displayed and the candidate can select a job before proceeding.	Pass
AT-03	Candidate Interface	Candidate uploads a CV and reviews extracted information.	The CV is accepted, extracted information is displayed, and the candidate can review it before submission.	Pass

Test ID	Application Area	Test Scenario	Expected Result	Status
AT-04	Candidate Interface	Candidate records video responses to screening questions.	Screening questions are displayed and the candidate can record video responses for the selected job.	Pass
AT-05	Candidate Interface	Candidate submits video responses for processing.	The video is uploaded successfully and the system provides upload feedback.	Pass
AT-06	Candidate Interface	Candidate completes the application.	A confirmation message is displayed after successful CV and video submission.	Pass
AT-07	Recruiter Dashboard	Recruiter logs in to the dashboard.	Valid credentials allow access to the dashboard, while invalid credentials are rejected.	Pass
AT-08	Recruiter Dashboard	Recruiter creates a job posting with job description and screening questions.	The job posting is saved and can be used for candidate applications.	Pass
AT-09	Recruiter Dashboard	Recruiter edits, activates, or deactivates a job posting.	Changes are saved successfully and job visibility updates according to status.	Pass
AT-10	Recruiter Dashboard	Recruiter views applicants, CV summaries, and video responses.	Applicant list is displayed with accessible CV information and video playback links.	Pass
AT-11	Recruiter Dashboard	Recruiter views AI-generated recommendation.	Candidate scores, recommendation, and analysis outputs are displayed correctly.	Pass
AT-12	Recruiter Dashboard	Recruiter accepts, rejects, or sends a candidate to a human interview.	The selected decision is saved and the applicant status is updated.	Pass
AT-13	Recruiter Dashboard	Recruiter reports a system issue.	The issue report is submitted and stored for admin-level review.	Pass
AT-14	Company Dashboard	Company user manages recruiter accounts.	Pending recruiters can be approved or removed according to company-level permissions.	Pass

Test ID	Application Area	Test Scenario	Expected Result	Status
AT-15	Company Dashboard	Company user views jobs and applicant overview.	The company dashboard displays job activity and applicant overview across the organization.	Pass
AT-16	Super Admin Dashboard	Super Admin views platform overview and manages companies.	Platform-level information is displayed and company accounts can be approved, rejected, or removed.	Pass
AT-17	Access Control	User attempts to access pages outside their role.	The system blocks unauthorized access and only displays features allowed for the logged-in role.	Pass
AT-18	Interface Responsiveness	Candidate and recruiter interfaces are tested under normal use.	Main pages and user actions respond within an acceptable time.	Pass
AT-19	AI Processing Feedback	A failed or uncertain AI processing case is tested.	The system does not discard the application and marks it for manual HR review where required.	Pass
AT-20	Recruiter Decision Flow	Recruiter overrides the AI recommendation.	The recruiter can make a final decision independently of the AI output, and the decision is saved.	Pass

The application testing results showed that the major user-facing workflows were functional. Candidates were able to move through the application process from job selection to CV upload, video response submission, and confirmation. Recruiters were able to create jobs, review applicants, view AI-generated results, and make final decisions. Company and Super Admin users were also able to perform their administrative actions, including recruiter approval, company management, and issue tracking.

The use of automated browser testing helped validate repeated workflows more efficiently. Selenium was useful for checking whether key interface actions continued to work after frontend or backend changes. Since these tests simulated real browser actions, they helped confirm that the application behaved correctly from the user's perspective. Overall, the application testing showed that NeuroHire is not only a set of independent AI modules, but a working recruitment platform with connected user workflows.

5.4 API Testing

API testing was conducted to verify that the NeuroHire backend could communicate correctly with the required processing services and return usable outputs to the application. The purpose of this testing was not to evaluate model accuracy, but to confirm that requests, responses, error handling, and data flow between the frontend, backend, and AI modules worked as expected. This included testing the FastAPI endpoints, CV-JD matching service, Whisper transcription and translation flow, OpenCV/MediaPipe-based video analysis, and LLM-based answer relevance output. The tests confirmed that the backend could send the required inputs to the deployed services, receive structured responses, and make those results available for candidate evaluation and recruiter review.

Table 5.3: API testing plan for NeuroHire

Test ID	**API Area**	**Test Scenario**	**Expected Result**	**Status**
API-01	Authentication	Send valid and invalid login credentials.	Valid users are authenticated, while invalid credentials are rejected.	Pass
API-02	CV-JD Matching	Send CV text and job description to the matching service.	The API returns a matching score and recommendation.	Pass
API-03	Whisper Processing	Send English and Urdu audio/video input for speech processing.	The API returns transcription or translation output successfully.	Pass
API-04	Video Analysis	Send candidate video input for behavioral analysis.	The API returns confidence and related video analysis outputs.	Pass
API-05	Answer Relevance	Send interview question and candidate transcript.	The API returns relevance, clarity, depth, and summary outputs.	Pass
API-06	Applicant Results	Request processed applicant data for recruiter dashboard.	Candidate scores, CV-JD result, video analysis result, and recommendation are returned correctly.	Pass
API-07	Error Handling	Send missing, incomplete, or invalid input to selected endpoints.	The API returns a controlled error response or marks the case for manual review.	Pass

6. Conclusion and Future Work

NeuroHire presents an AI-driven approach to high-volume recruitment by integrating multimodal candidate evaluation through CV analysis and video-based assessment. The system enhances efficiency, consistency, and fairness in screening while supporting informed decision-making through structured scoring and recommendations. By combining automation with human oversight, NeuroHire provides a scalable and practical solution for modern recruitment challenges. Future enhancements include enabling real-time feedback for candidates, expanding support for additional regional languages, integrating with enterprise recruitment systems for broader adoption, and developing and deploying an AI-powered interview booth to provide a controlled and standardized assessment environment.

7. Reflection

7.1 Learning Reflection

The development of NeuroHire provided hands-on experience in building an end-to-end AI system integrating multiple modalities, including CV analysis, speech processing, and behavioral video analysis. We gained practical skills in working with large language models (LLMs), designing evaluation frameworks using OpenAI Evals, and handling real-world data from recruitment settings. Additionally, we learned how to design explainable AI systems that support decision-making rather than replace it. One key challenge was ensuring consistency across different modules, especially when integrating outputs from CV–JD matching and speech analysis. Overcoming this required iterative testing, debugging, and refining evaluation strategies.

7.2 Team Reflection

The project required strong collaboration across different technical components such as backend development, AI model integration, frontend design, and evaluation. Responsibilities were divided based on individual strengths, with team members focusing on areas like LLM development, system integration, UI design, and data processing. Regular communication and coordination were essential to ensure smooth integration of modules. One challenge was aligning different components into a unified pipeline, which required frequent discussions and iterative adjustments. Overall, teamwork improved our ability to manage a complex system collaboratively.

7.3 Project Reflection

Initially, NeuroHire was proposed as an AI-based recruitment screening system focusing on CV analysis and video interviews. Over time, the project evolved into a more comprehensive system with the addition of LLM-based answer relevance analysis, OpenAI Evals for structured evaluation, and bilingual English–Urdu support. Some features were refined or simplified to ensure feasibility within the given timeframe, while others, such as explainable scoring and real-world evaluation using KFC data, were added to strengthen practical relevance. These changes were driven by challenges in implementation, availability of data, and the need to focus on impactful features. The project highlights opportunities for scalable hiring solutions while also revealing challenges in handling real-world variability and ensuring fairness in AI-assisted recruitment.

7.4 Process Reflection

The iterative development approach worked well, allowing continuous improvement of individual modules and their integration into the overall system. Testing with real-world data and evaluating performance using structured metrics helped validate system effectiveness. However, the process could be improved by incorporating earlier end-to-end integration and more extensive dataset collection from the beginning. Future improvements could include better planning for data acquisition, more robust evaluation pipelines, and enhanced user testing to further refine the system.

Appendix A. Data

A.1 Dataset Links

The following datasets were used for evaluating different modules of NeuroHire:

- **HR Interview Questions Dataset**
  Used for constructing interview question scenarios for the answer relevance evaluation.
  Kaggle HR Interview Questions and Ideal Answers Dataset

- **Mozilla Common Voice Dataset**
  Used for evaluating English speech-to-text transcription using Whisper.
  Mozilla Common Voice Dataset

- **Urdu Speech Dataset**
  Used for evaluating Urdu-to-English translation using Whisper.
  Urdu Speech Dataset

- **KFC Recruitment Data**
  We also collected real-world recruitment data from KFC in the form of candidate resumes and video interviews. This data was used for internal evaluation of CV parsing, CV–JD matching, and video interview analysis modules. However, it cannot be publicly shared due to confidentiality and privacy reasons.

Appendix B. Code

The NeuroHire source code and project materials are available at:
https://github.com/is07353/neurohire-fyp

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